

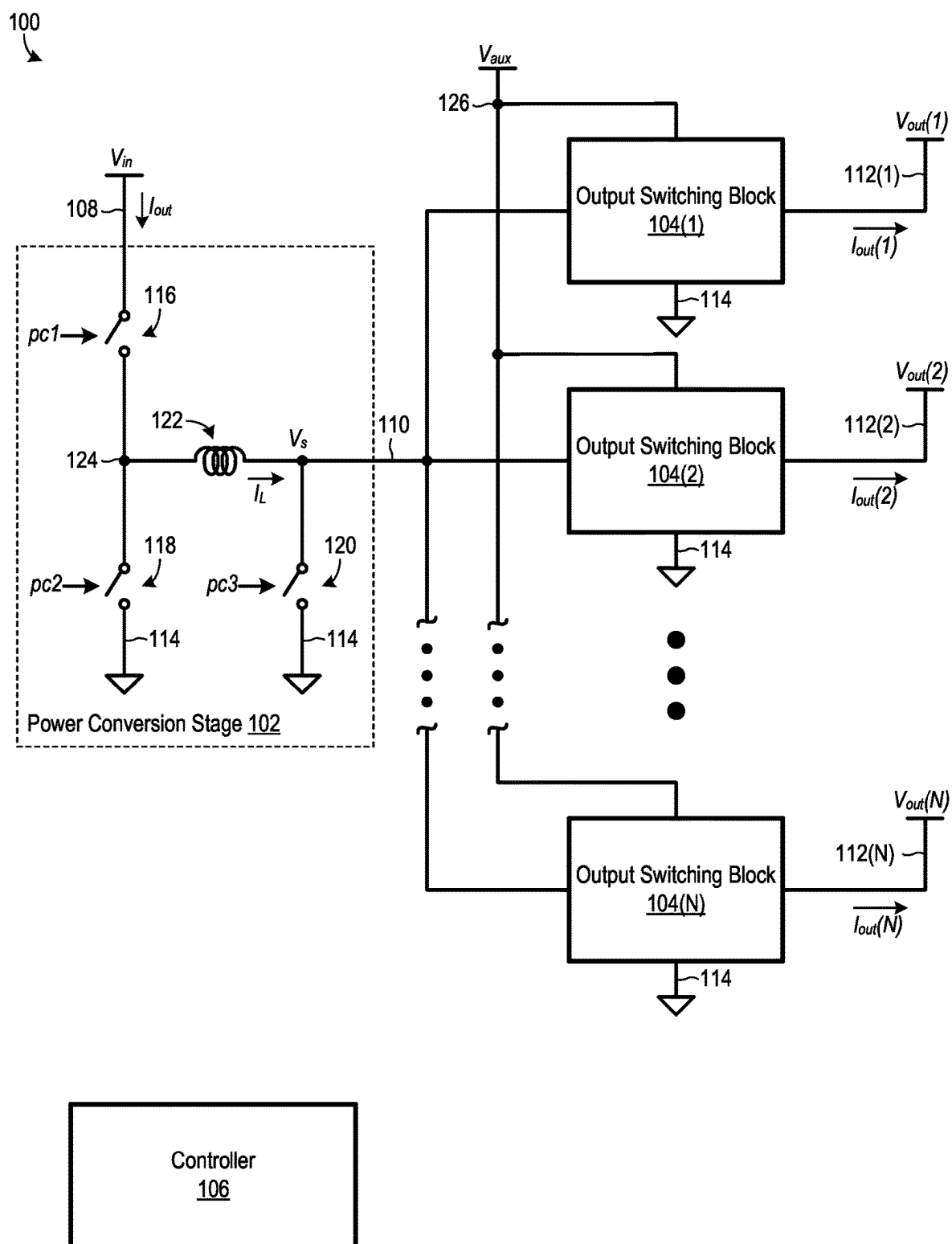
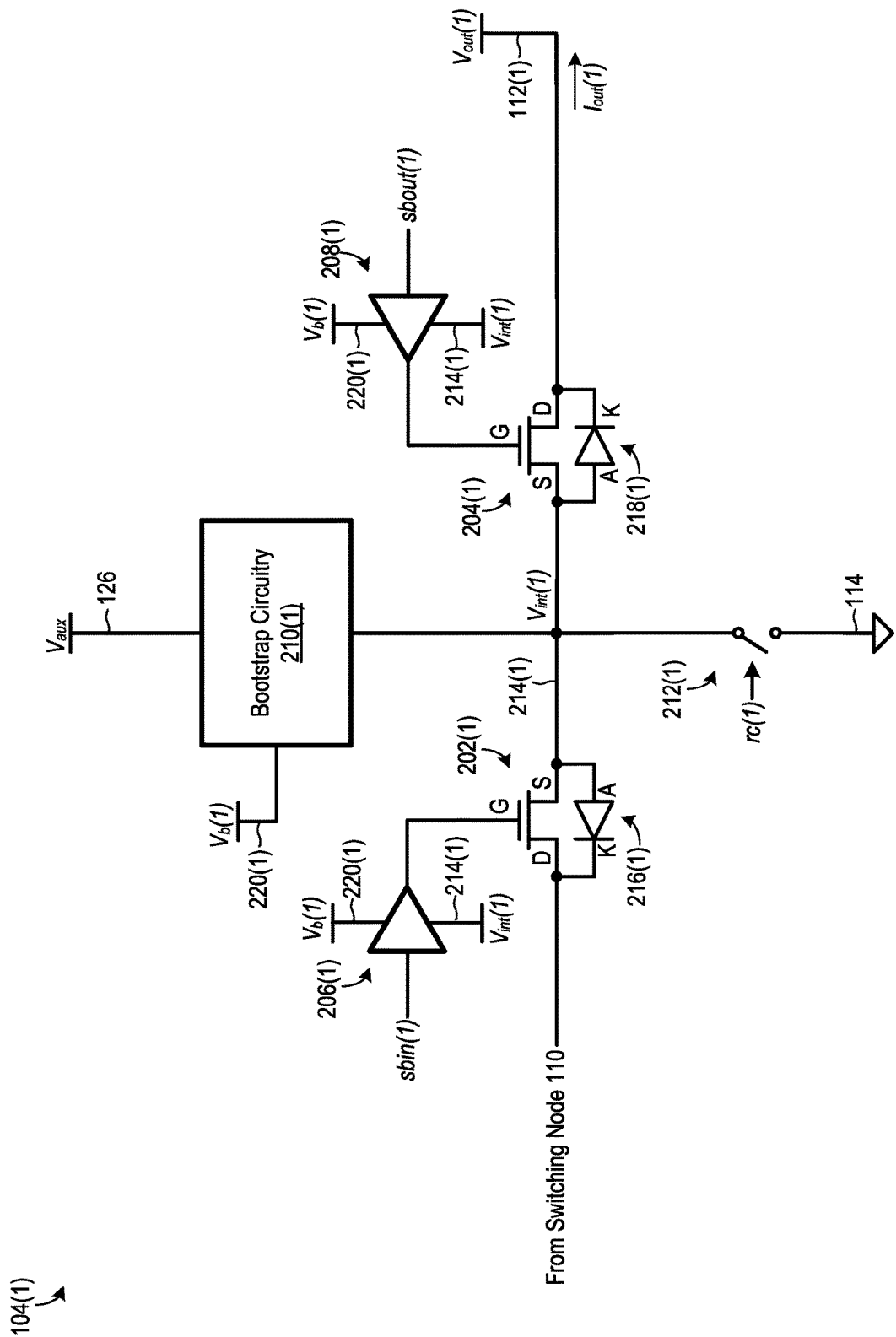
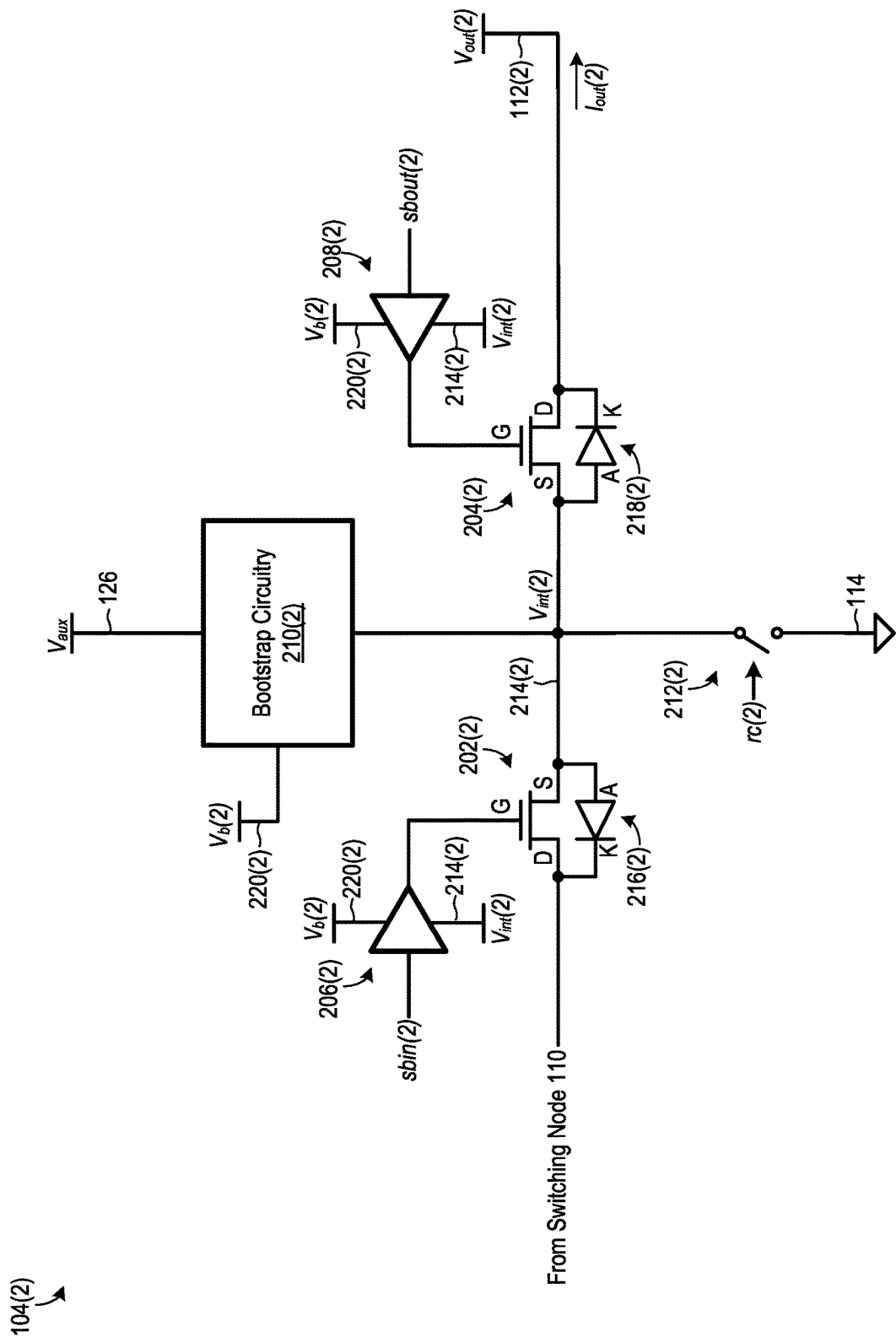


FIG. 1





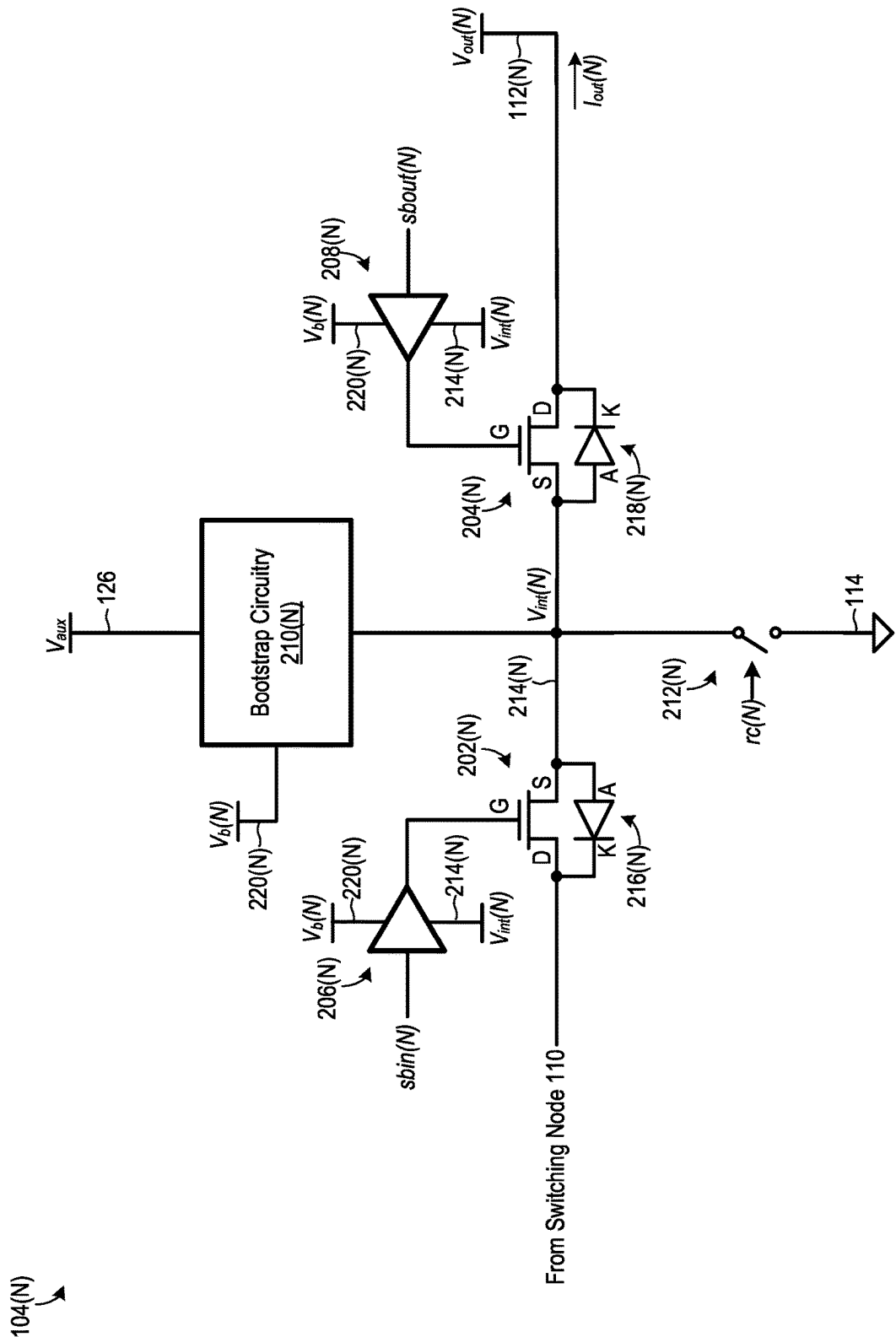
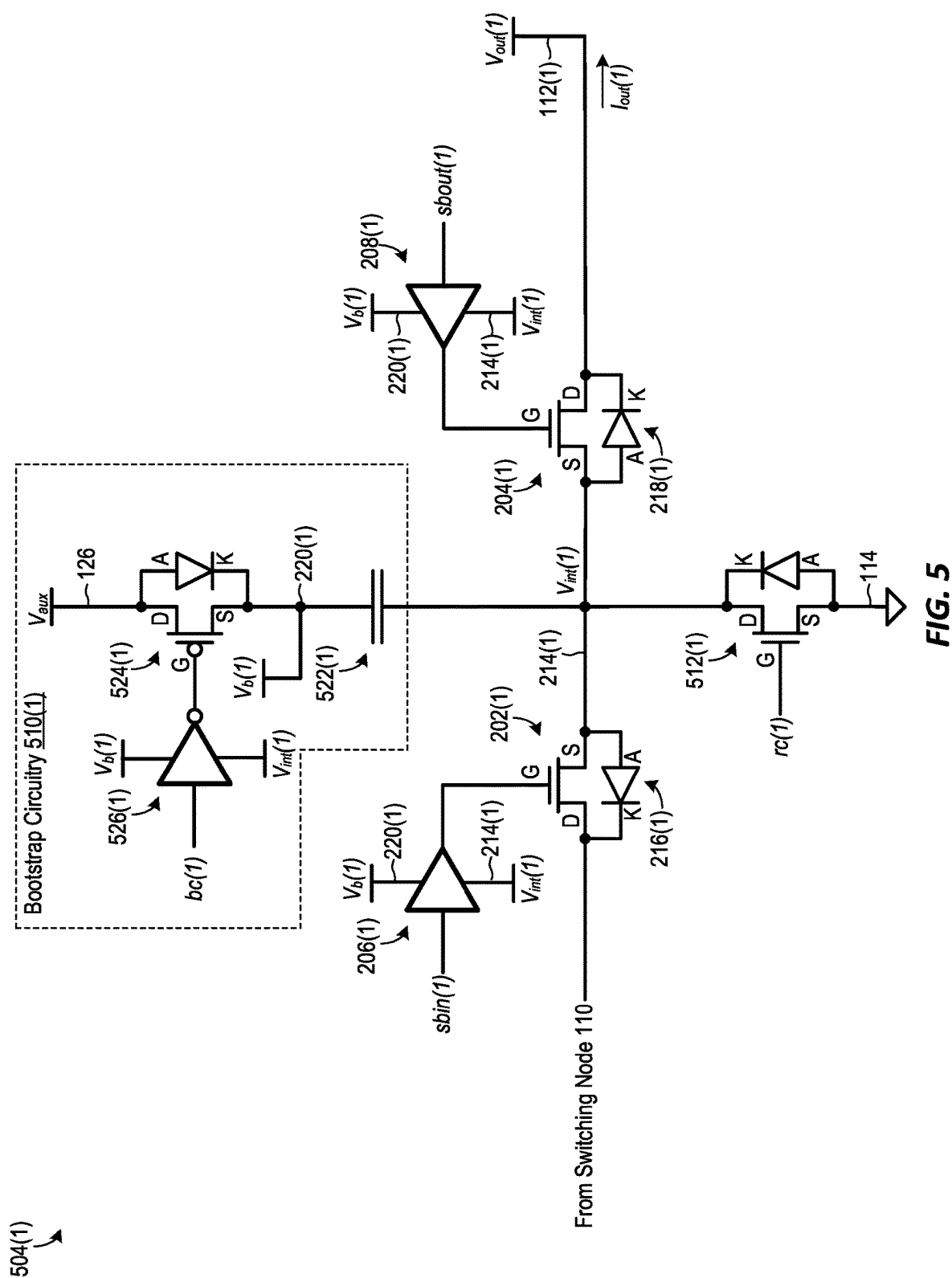
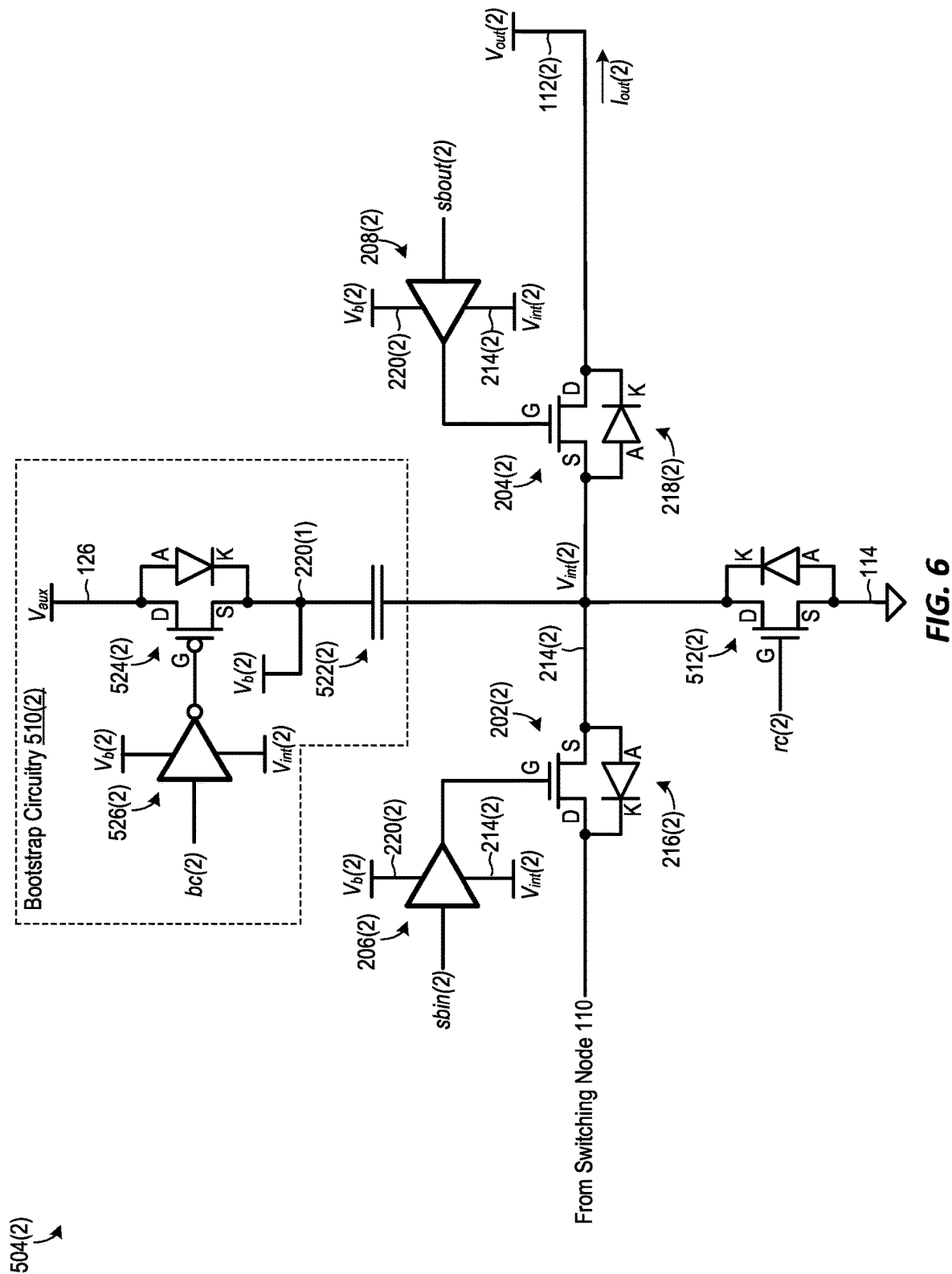
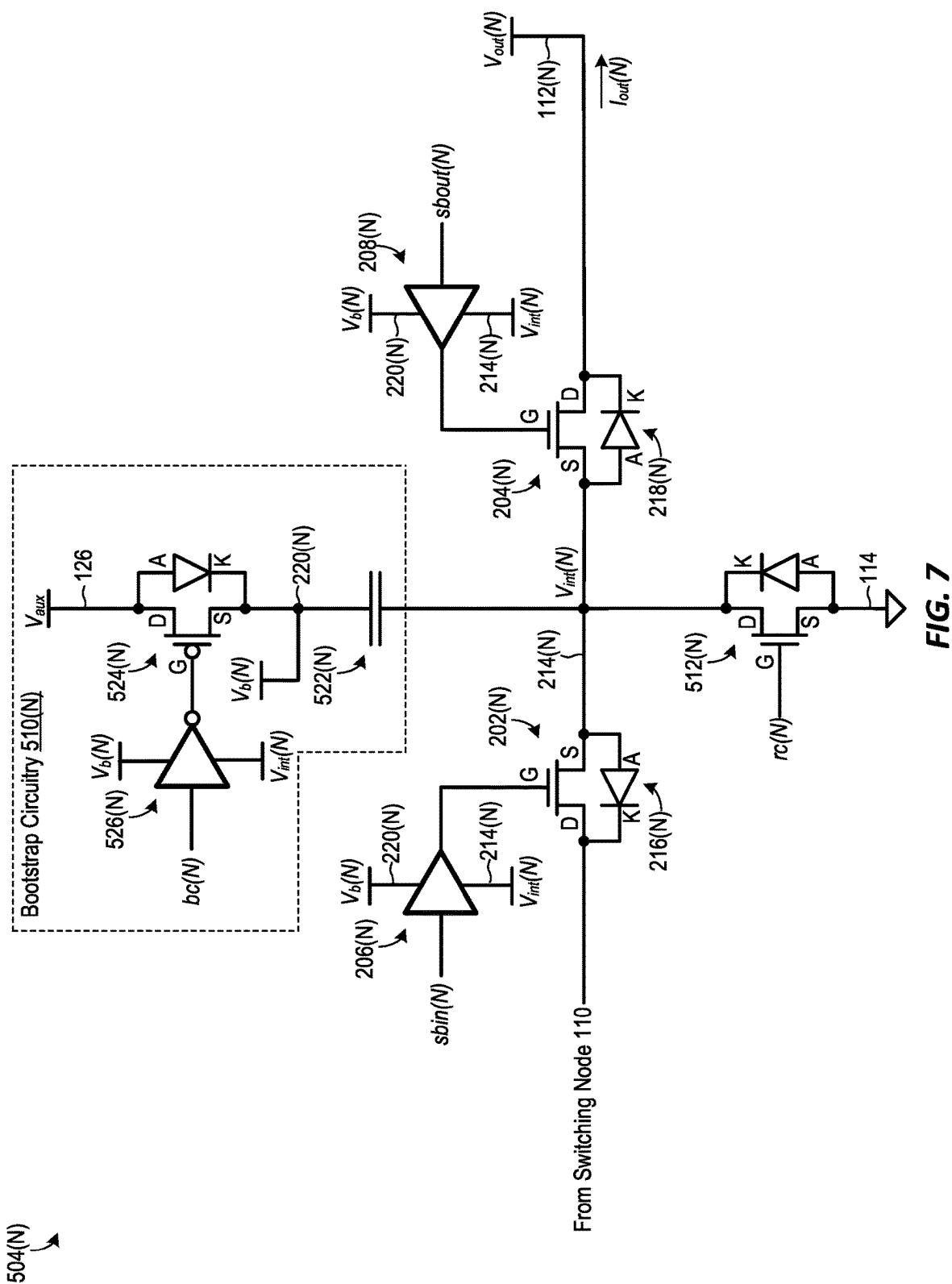


FIG. 4







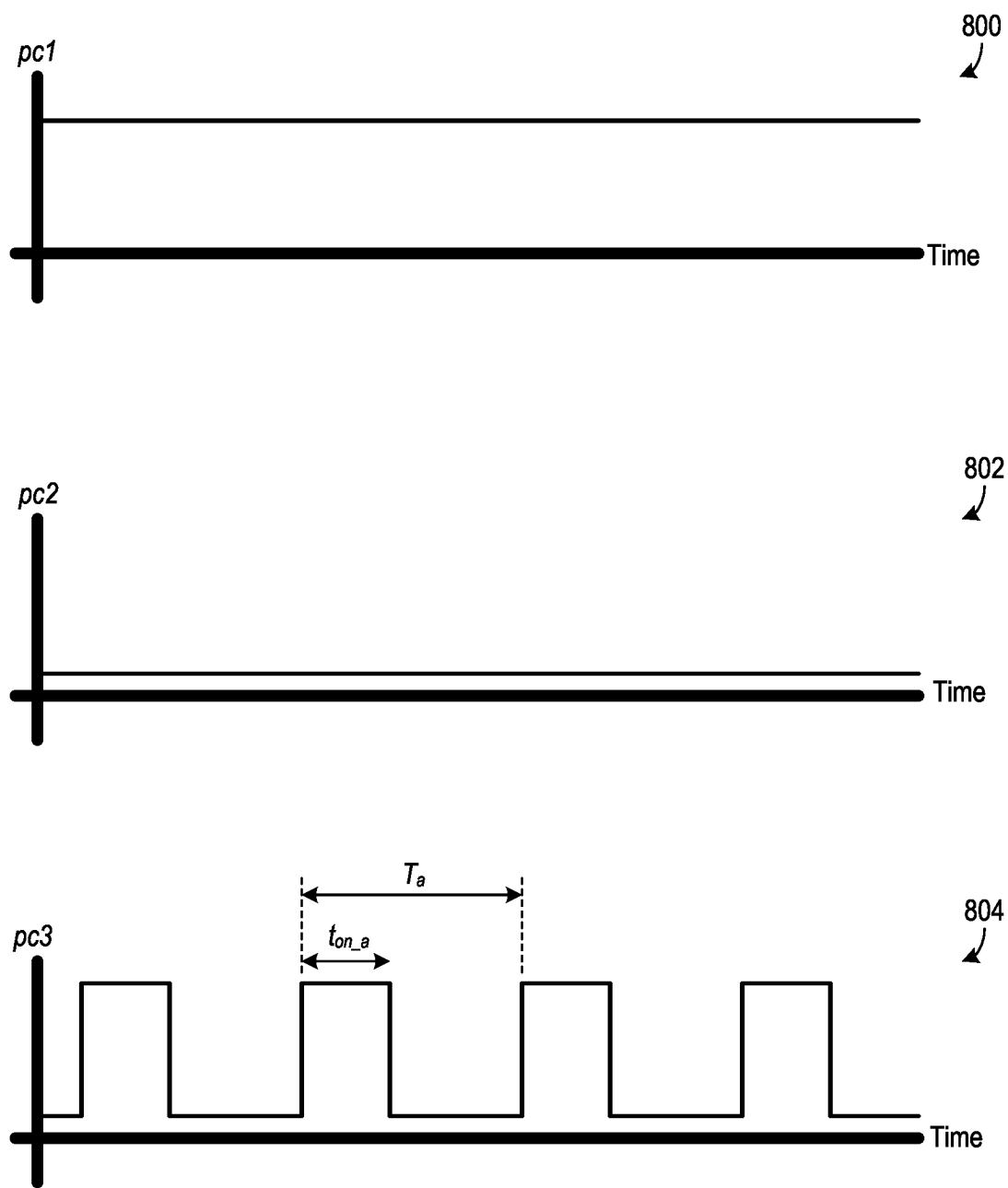


FIG. 8

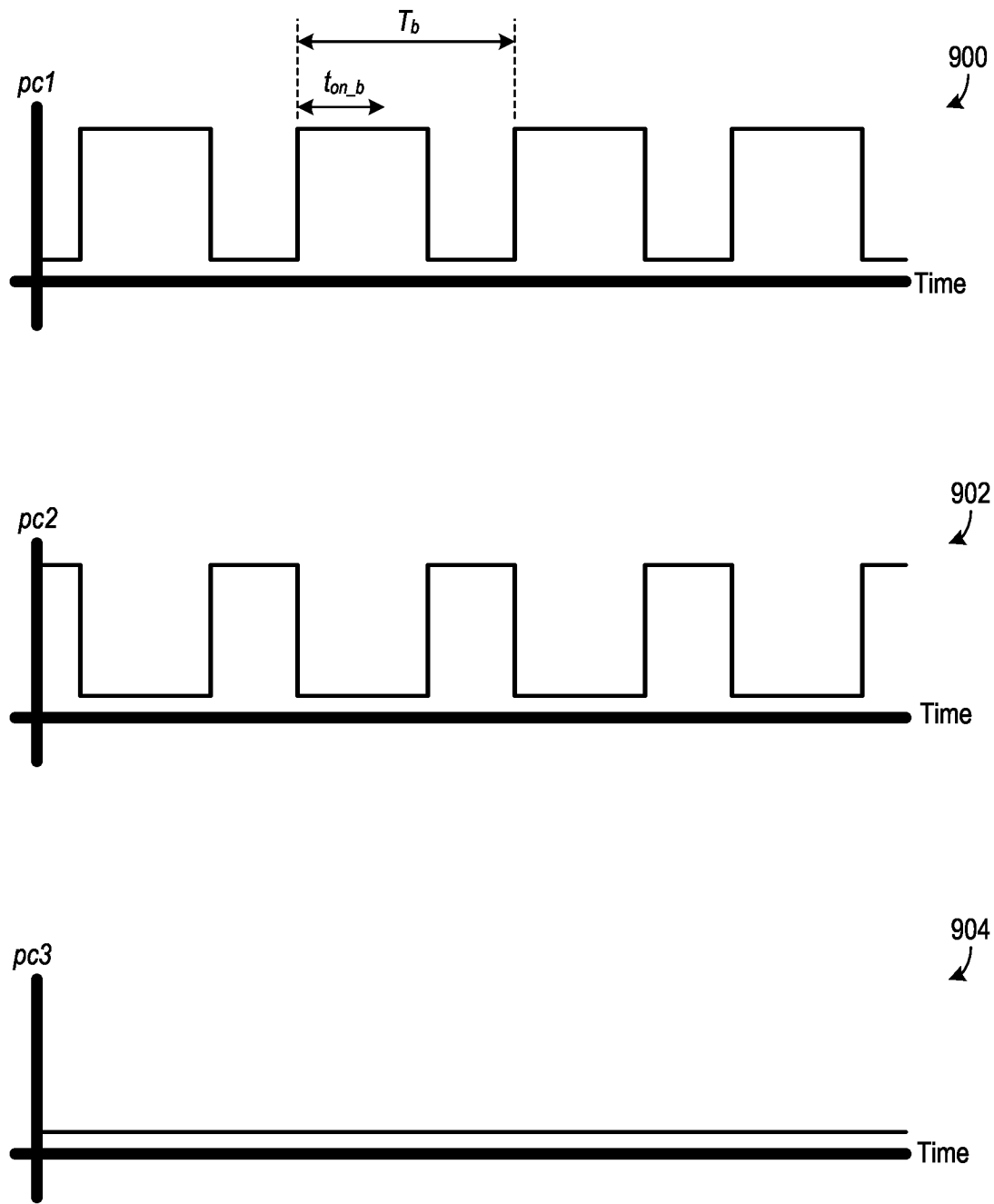


FIG. 9

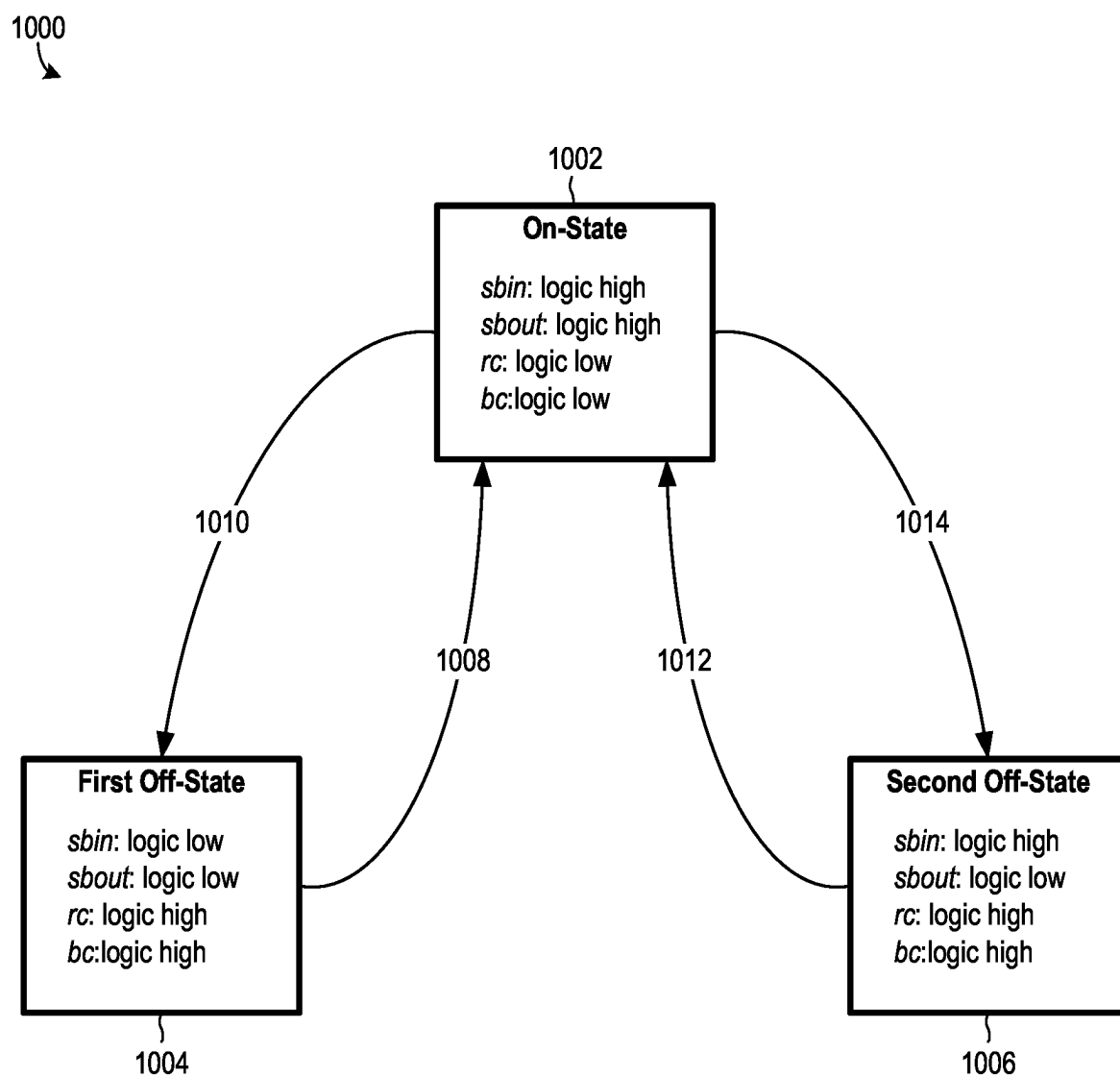


FIG. 10

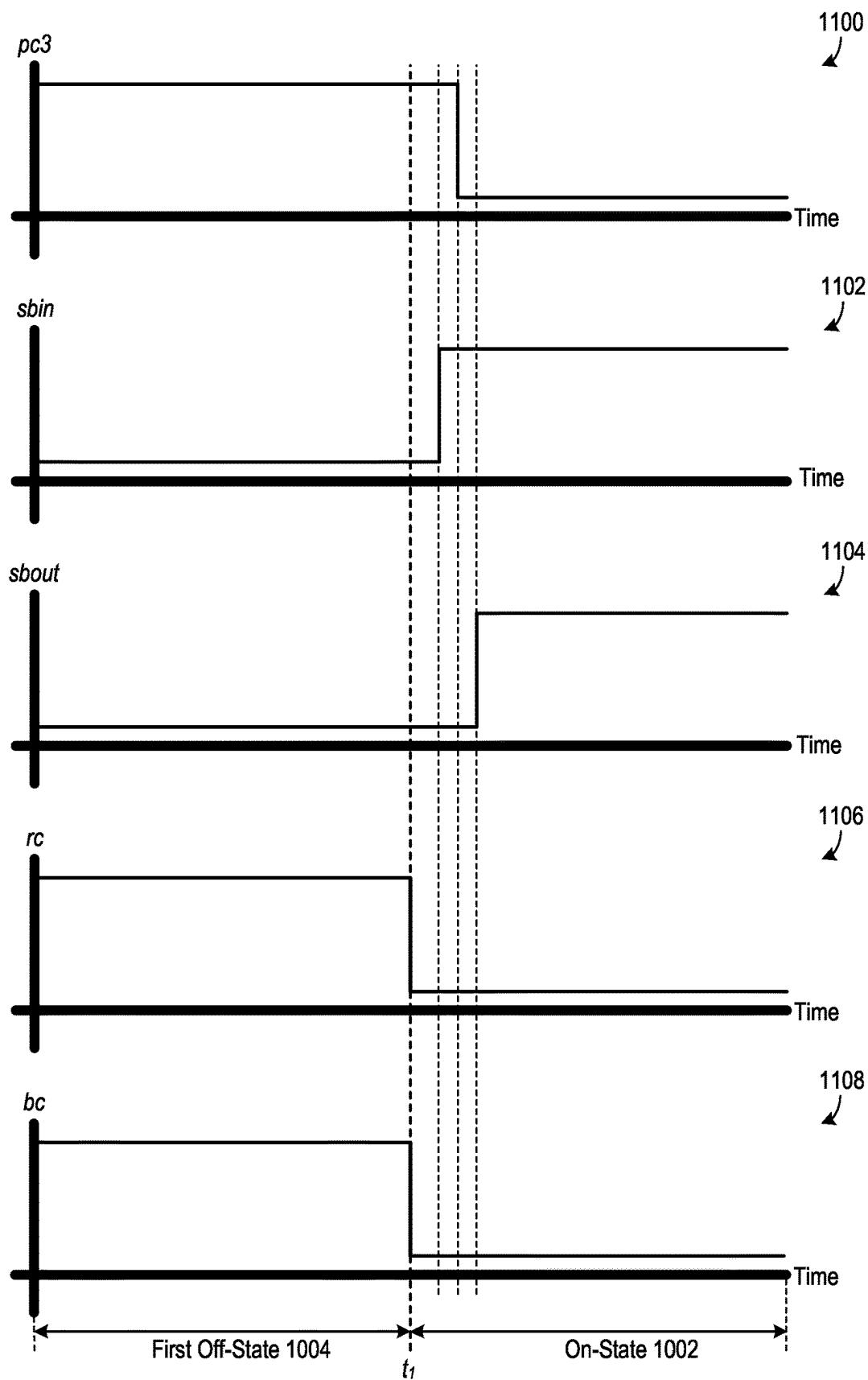


FIG. 11

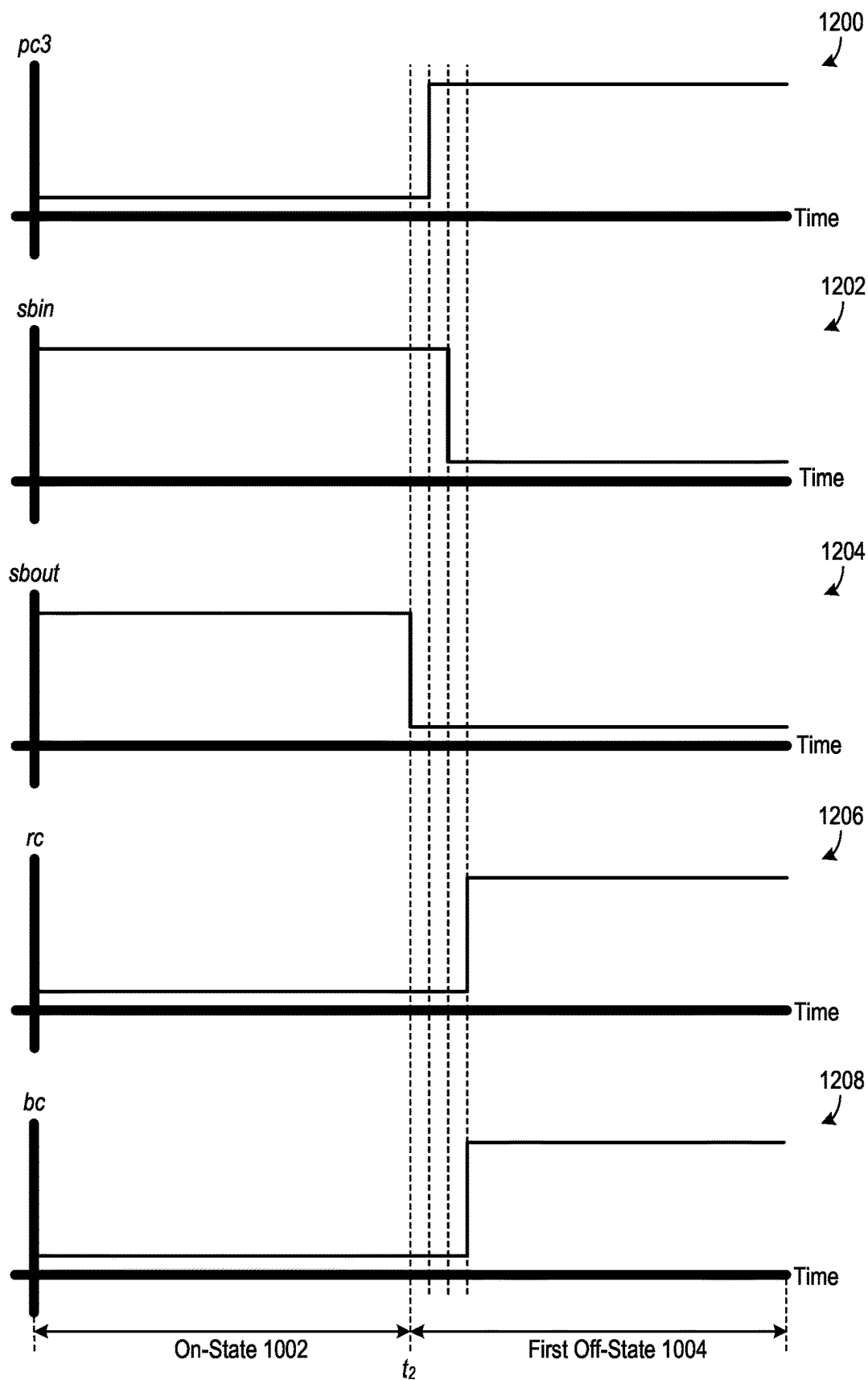


FIG. 12

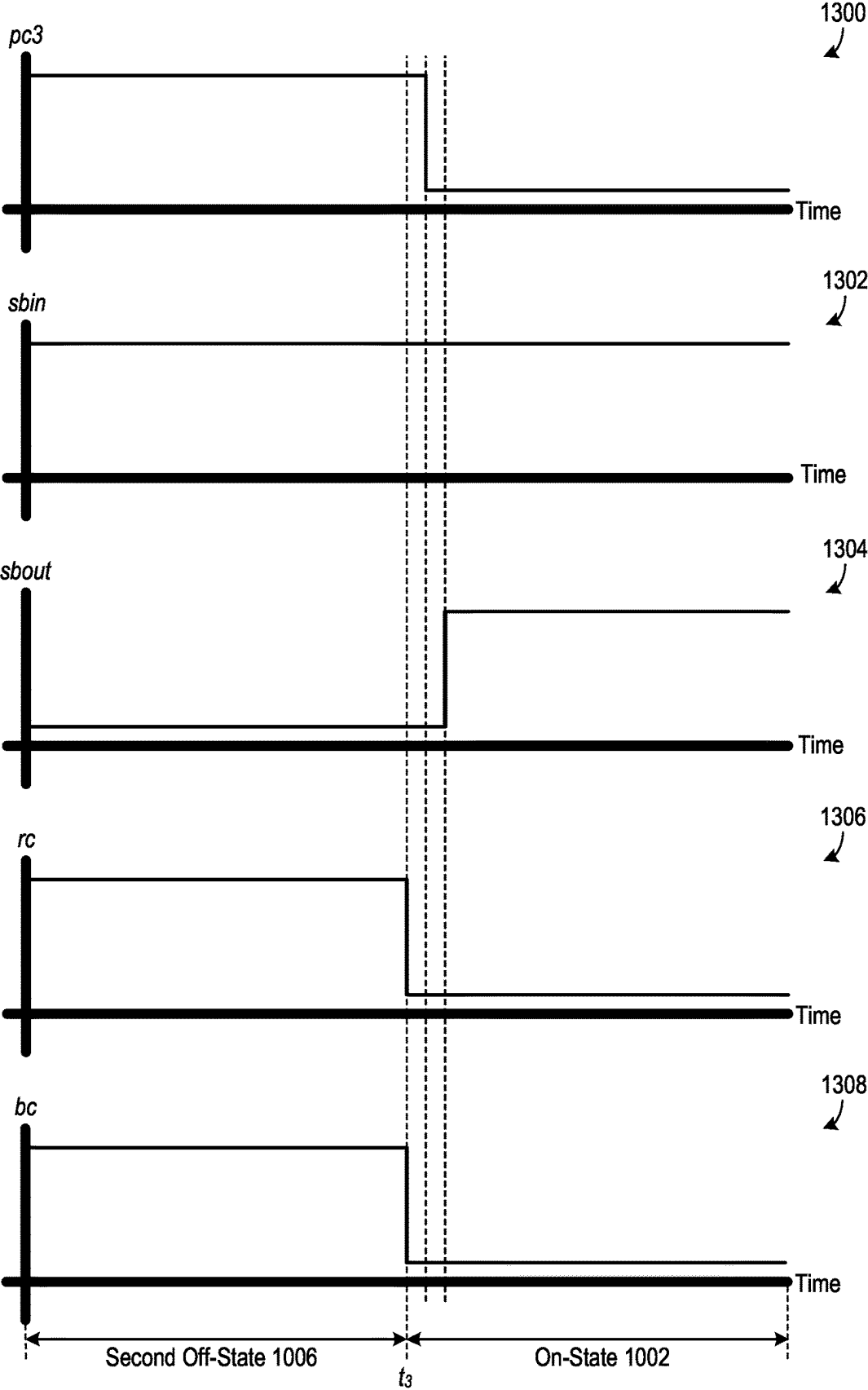


FIG. 13

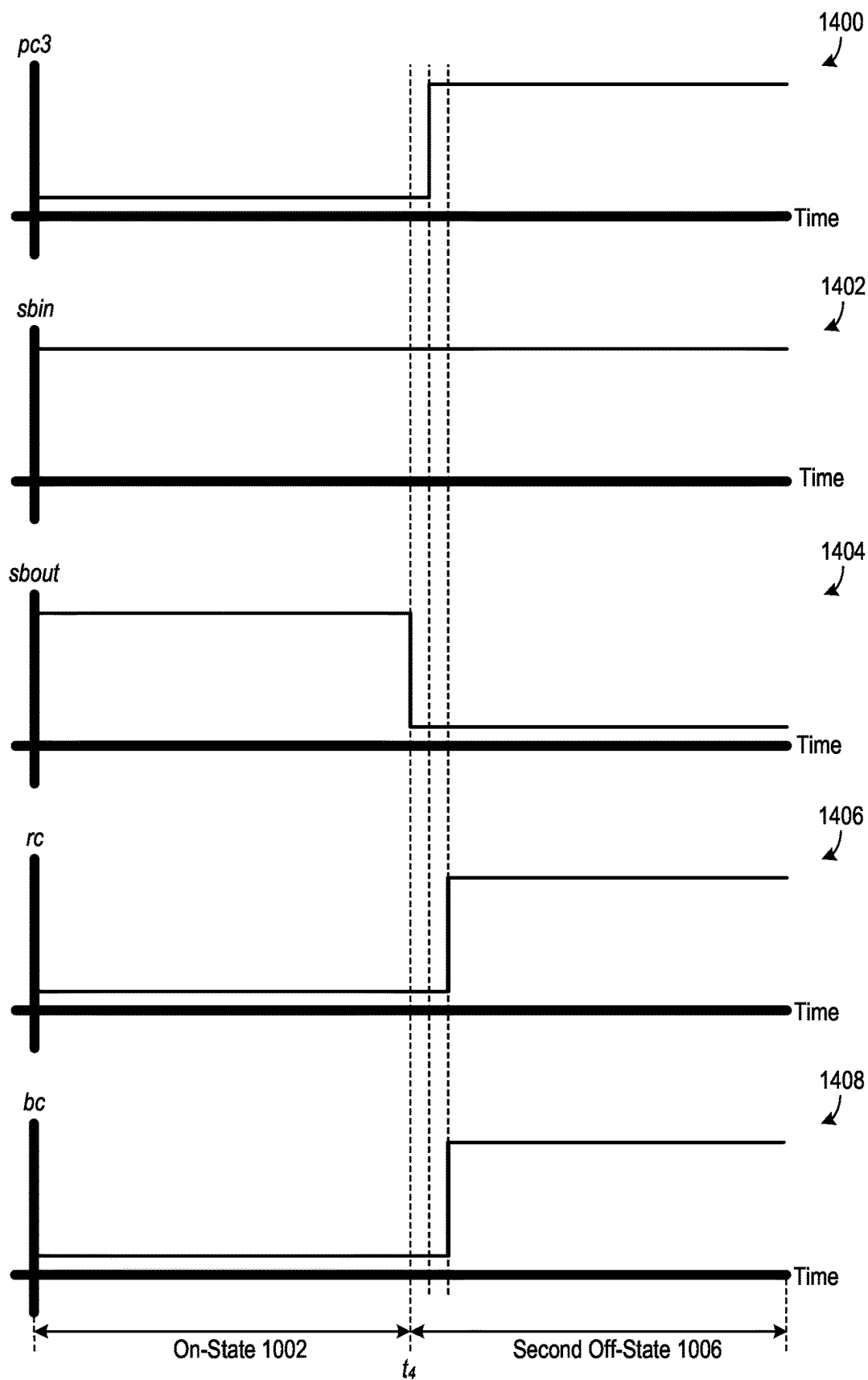
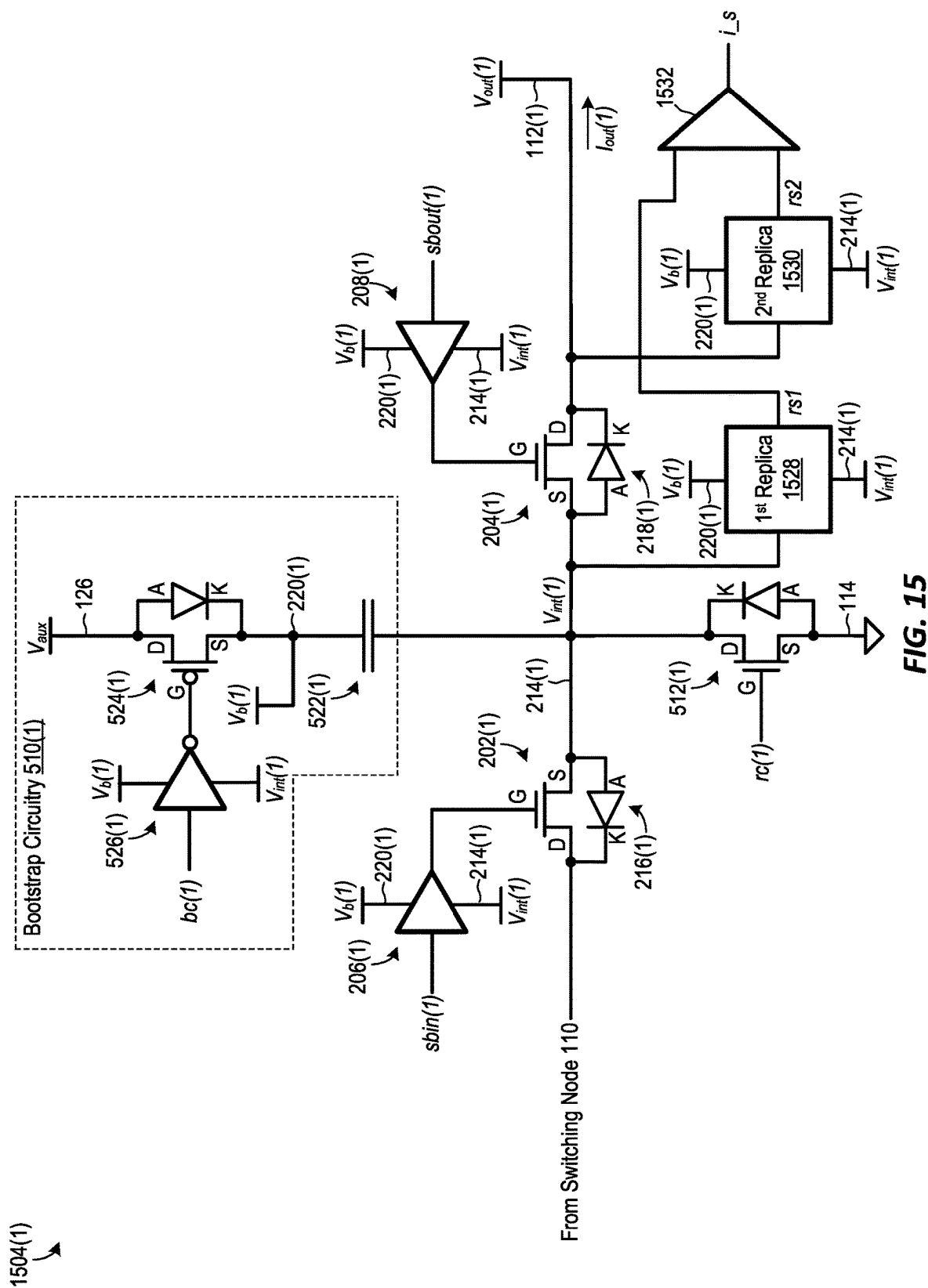


FIG. 14



SINGLE-INPUT, MULTI-OUTPUT DIRECT-CURRENT-TO-DIRECT CURRENT CONVERTERS AND ASSOCIATED METHODS

BACKGROUND

[0001] Many electrical devices require multiple power supply rails. For example, an electrical device may require two or more power supply rails to power circuitry having different power supply voltage requirements. As another example, an electrical device may require two or more power supply rails to enable two or more systems within the electrical device to be independently powered on and powered off.

BRIEF DESCRIPTION OF THE DRAWINGS

[0002] FIG. 1 is a schematic diagram of a single-input, multiple-output (SIMO) direct-current-to-direct-current (DC-to-DC) converter, according to an embodiment.

[0003] FIG. 2 is a schematic diagram of a first output switching block of the FIG. 1 SIMO DC-to-DC converter.

[0004] FIG. 3 is a schematic diagram of a second output switching block of the FIG. 1 SIMO DC-to-DC converter.

[0005] FIG. 4 is a schematic diagram of an Nth output switching block of the FIG. 1 SIMO DC-to-DC converter.

[0006] FIG. 5 is a schematic diagram of one embodiment of the first output switching block of FIG. 2.

[0007] FIG. 6 is a schematic diagram of one embodiment of the second output switching block of FIG. 3.

[0008] FIG. 7 is a schematic diagram of one embodiment of the Nth output switching block of FIG. 4.

[0009] FIG. 8 includes three graphs collectively illustrating example control signals for a power conversion stage of the FIG. 1 SIMO DC-to-DC converter when the power conversion stage operates as a boost converter.

[0010] FIG. 9 includes three graphs collectively illustrating example control signals for a power conversion stage of the FIG. 1 SIMO DC-to-DC converter when the power conversion stage operates as a buck converter.

[0011] FIG. 10 is a state diagram illustrating example operating states of the FIG. 1 SIMO DC-to-DC converter.

[0012] FIG. 11 includes five graphs collectively illustrating an example of an output switching block transitioning from a first off-state to an on-state.

[0013] FIG. 12 includes five graphs collectively illustrating an example of an output switching block transitioning from the on-state to the first off-state.

[0014] FIG. 13 includes five graphs collectively illustrating an example of an output switching block transitioning from a second off-state to the on-state.

[0015] FIG. 14 includes five graphs collectively illustrating an example of an output switching block transitioning from the on-state to the second off-state.

[0016] FIG. 15 is a schematic diagram of an alternate embodiment of the FIG. 5 output switching block that further includes elements to perform current sensing.

DETAILED DESCRIPTION OF THE EMBODIMENTS

[0017] A plurality of power supply rails in an electrical device can be provided by providing a respective power converter, such as a respective direct-current-to-direct-current (DC-to-DC) converter, for each power supply rail.

However, providing a respective power converter for each power supply rail may be expensive and require significant space. Therefore, it is frequently desirable to provide multiple power supply rails from a single power converter. For example, a single-input, multi-output (SIMO) DC-to-DC converter may be used to provide multiple output power nodes from a single input power node. The output power nodes, though, require output switching circuitry with bidirectional current blocking capability to ensure that each output power node is electrically isolated from each other output power node, as well as to enable control of which output power node is powered by the SIMO DC-to-DC converter at a given time.

[0018] It is frequently desirable to implement a switch, such as in output switching circuitry of a SIMO DC-to-DC converter, using an enhancement mode, N-type metal oxide semiconductor field effect transistor (NMOS FET) because an NMOS FET generally has lower on-resistance than an enhancement mode, P-type metal oxide semiconductor field effect transistor (PMOS FET) of similar size. Additionally, NMOS FET fabrication is highly developed in modern semiconductor fabrication processes. However, an NMOS FET includes a body diode between its source and drain which generally prevents the NMOS FET from blocking bidirectional current. While this issue can be overcome by managing the body diode through body switching, body switching is risky and non-robust because the body is floating for a short time during switching, which impairs breakdown voltage rating of the NMOS FET. Additionally, body switching necessitates complex gate drive schemes, and there may be limited support for body switching in a semiconductor fabrication process.

[0019] As such, it is generally desirable to electrically couple two NMOS FETs in series with opposing body diode orientations to enable bidirectional current blocking, instead of implementing body switching. However, gate drive of two series-connected NMOS FETs in output switching circuitry of a SIMO DC-to-DC converter is non-trivial. For example, one conventional approach for driving gates of NMOS FETs in output switching circuitry of a SIMO DC-to-DC converter requires a respective bootstrap driving domain referred to the switching node and a respective power supply rail for each output power node referred to the output power node itself, which may be costly, space consuming, and quiescent current demanding. As another example, another conventional approach for driving gates of NMOS FETs in output switching circuitry of a SIMO DC-to-DC converter relies on a single power supply rail to power high-side NMOS FETs across multiple output power nodes, which may limit operating voltage range and/or be unable to achieve high performance.

[0020] Disclosed herein are new SIMO DC-to-DC converters which at least partially overcome the above discussed drawbacks. The new SIMO DC-to-DC converters include a respective output switching block for each output power node where each output switching block requires only a single respective power supply rail, thereby promoting low SIMO DC-to-DC converter cost and small SIMO DC-to-DC converter size. Additionally, NMOS FETs are oriented in the output switching blocks of the new SIMO DC-to-DC converters in a manner that promotes robust and simple gate drive of the NMOS FETs with a single power supply rail, thereby helping achieve high performance, such as low on-resistance. Furthermore, the new SIMO DC-to-DC con-

verters do not require field effect transistor body switching, thereby further promoting robust operation. Moreover, particular embodiments of the new SIMO DC-to-DC converters can achieve high performance operation over a wider voltage range than can be realized by typical conventional SIMO DC-to-DC converters. For example, certain embodiments are easily scalable to operate over different voltage ranges simply by changing the drain-to-source and drain-to-gate voltage ratings of constituent NMOS FETs, which are frequently available in a wide range of ratings in typical semiconductor fabrication processes.

[0021] FIG. 1 is a schematic diagram of a SIMO DC-to-DC converter 100, which is one embodiment of the new SIMO DC-to-DC converters disclosed herein. The SIMO DC-to-DC converter 100 includes a power conversion stage 102, N output switching blocks 104, and a controller 106, where N is an integer greater than or equal to two. In this document, specific instances of an item may be referred to by use of a numeral in parentheses (e.g. the output switching block 104(1)) while numerals without parentheses refer to any such item (e.g. the output switching blocks 104). Although FIG. 1 illustrates N being at least three, it is understood that N could be equal to two. The power conversion stage 102 is electrically coupled between an input power node 108 and a switching node 110, and each output switching block 104 is electrically coupled between the switching node 110 and a respective output power node 112. Each output power node implements, for example, a respective power supply rail that electrically powers one or more loads (not shown). The SIMO DC-to-DC converter 100 can, and typically will, include additional components, such as one or more capacitors (not shown), electrostatic discharge (ESD) protection circuitry (not shown), etc.

[0022] The input power node 108 has an input voltage V_{in} with respect to a reference node 114, and each output power node 112 has a respective output voltage V_{out} with respect to the reference node 114. While not required, in some embodiments, nominal magnitude of the output voltage V_{out} varies among two or more of the output power nodes 112, such as to enable the SIMO DC-to-DC converter 100 to power two or more loads requiring different power supply voltage magnitude. The reference node 114 is depicted as being a ground node, such as an earth ground node or a chassis ground node. It is understood, though, that the reference node 114 need not be a ground node, and the reference node 114 accordingly could be at a different electrical potential than an earth ground or a chassis ground. An input current I_{in} flows into the SIMO DC-to-DC converter 100 from the input power node 108, and a respective output current I_{out} flows out of each output power node 112 to a load (not shown) electrically coupled to the output power node 112. A respective polarity of the input current I_{in} and each output current I_{out} could be either positive or negative, depending on the operating conditions of the SIMO DC-to-DC converter 100.

[0023] The power conversion stage 102 includes a power conversion switching device 116, a power conversion switching device 118, a power conversion switching device 120, and an inductor 122. Each of the aforementioned power conversion switching devices includes, for example, one or more transistors and associated driver circuitry. The power conversion switching device 116 is electrically coupled between the input power node 108 and a connection node 124, and the power conversion switching device 118 is

electrically connected between the connection node 124 and the reference node 114. The inductor 122 is electrically coupled between the connection node 124 and the switching node 110, and the power conversion switching device 120 is electrically coupled between the switching node 110 and the reference node 114. The power conversion switching device 116 is controlled by a signal pc1 generated by the controller 106, the power conversion switching device 118 is controlled by a signal pc2 generated by the controller 106, and the power conversion switching device 120 is controlled by a signal pc3 generated by the controller 106. As discussed below, the controller 106 generates the control signal pc1, the control signal pc2, and/or the control signal pc3 to cause the power conversion stage 102 to provide a regulated voltage V_s with respect to the reference node 114 at the switching node 110.

[0024] As discussed below, each output switching block 104 is configured to connect and disconnect its respective output power node 112 from the switching node 110 under the command of the controller 106. Additionally, each output switching block 104 is capable of bidirectional current blocking and thereby electrically isolating its respective output power node 112 from each other output power node 112. Each output switching block 104 is at least partially electrically powered by an auxiliary power supply rail 126 having a voltage V_{aux} . Each output switching block 104 is also electrically coupled to the reference node 114. In some embodiments, each output switching block 104 is electrically powered from the input power node 108 instead of from the auxiliary power supply rail 126, and the auxiliary power supply rail 126 is therefore optionally omitted. Details of the output switching blocks 104 are not shown in FIG. 1, but FIGS. 2, 3, and 4, illustrate details of the output switching blocks 104(1), 104(2), and 104(N), respectively. It is understood that other instances of the output switching blocks 104 are embodied in a manner analogous to the output switching blocks 104(1), 104(2), and 104(N).

[0025] Referring to FIGS. 2-4, each output switching block 104 includes an input enhancement mode, N-type metal oxide semiconductor field effect transistor (input NMOS FET) 202, an output enhancement mode, N-type metal oxide semiconductor field effect transistor (output NMOS FET) 204, input switch driver circuitry 206, output switch driver circuitry 208, bootstrap circuitry 210, and a reference switching device 212. Each input NMOS FET 202 and each output NMOS FET 204 has a respective gate (G), a respective drain (D), and a respective source(S). Additionally, each input NMOS FET 202 and each output NMOS FET 204 includes a respective body diode 216 and 218 with an anode (A) and a cathode (K). In each output switching block 104, the input NMOS FET 202 is electrically coupled between the switching node 110 and an internal node 214 of the output switching block 104 with an orientation such that (i) the drain of the input NMOS FET 202, as well as the cathode of the body diode 216, are electrically coupled to the switching node 110, and (ii) the source of the input NMOS FET 202, as well as the anode of the body diode 216, are electrically coupled to the internal node 214 of the output switching block 104. Additionally, in each output switching block 104, the output NMOS FET 204 is electrically coupled between the internal node 214 of the output switching block 104 and the output power node 112 of the output switching block 104 with an orientation such that (i) the drain of the output NMOS FET 204, as well as the cathode of the body

diode **218**, are electrically coupled to the output power node **112**, and (ii) the source of the output NMOS FET **204**, as well as the anode of the body diode **218**, are electrically coupled to the internal node **214** of the output switching block **104**.

[0026] The bootstrap circuitry **210** of each output switching block **104** generates a respective bootstrap power supply rail **220** of the output switching block **104** having a voltage V_b that is referenced to the internal node **214** of the output switching block **104**. The input switch driver circuitry **206** of each output switching block **104** drives the gate of the input NMOS FET **202** of the output switching block **104** in response to a respective control signal s_{bin} generated by the controller **106** (FIG. 1). Additionally, the output switch driver circuitry **208** of each output switching block **104** drives the gate of the output NMOS FET **204** of the output switching block **104** in response to a respective control signal s_{out} generated by the controller **106** (FIG. 1). In each output switching block **104**, the input switch driver circuitry **206** and the output switch driver circuitry **208** are (i) powered from the bootstrap power supply rail **220** of the output switching block **104** and (ii) electrically reference to the internal node **214** of the output switching block **104** (instead of to the reference node **114**). As such, in each output switching block **104**, the input switch driver circuitry **206** drives the gate of the input NMOS FET **202** between the voltage V_b of the bootstrap power supply rail **220** and the voltage V_{int} of the internal node **214**, and the output switch driver circuitry **208** drives the gate of the output NMOS FET **204** between the voltage V_b of the bootstrap power supply rail **220** and the voltage V_{int} of the internal node **214**. The reference switching device **212** of each output switching block **104** is electrically coupled between the internal node **214** of the output switching block **104** and the reference node **114**.

[0027] The above-described orientations of the input NMOS FETs **202** and the output NMOS FETs **204** are important. In particular, not only do the orientations enable bidirectional current blocking by preventing current flow through an output switching block **104** solely via the body diodes **216** and **218**, but the orientations also facilitate gate drive of the NMOS FETs. Specifically, the above-described orientations of the input NMOS FET **202** and the output NMOS FET **204** in an output switching block **104** result in the source of each of the two FETs to be connected to a common node, i.e., the internal node **214** of the output switching block **104**, which enables the respective gates of the two FETs to receive a common gate-to-source voltage when being driven by a common power supply rail, i.e. bootstrap power supply rail **220**, which facilitates robust driving of the two FETs from the common power supply rail.

[0028] FIGS. 5-7 illustrate one possible embodiment of bootstrap circuitry **210** and reference switching device **212** of output switching blocks **104**. Specifically, FIG. 5 is a schematic diagram of an output switching block **504(1)**, which is an embodiment of output switching block **104(1)** (FIG. 2), FIG. 6 is a schematic diagram of an output switching block **504(2)**, which is an embodiment of output switching block **104(2)** (FIG. 3), and FIG. 7 is a schematic diagram of an output switching block **504(N)**, which is an embodiment of output switching block **104(N)** (FIG. 4). In the output switching blocks of FIGS. 5-7, bootstrap circuitry **210** is embodied by bootstrap circuitry **510**, and the reference switching device **212** is embodied by a reference

enhancement mode, N-type metal oxide semiconductor field effect transistor (reference NMOS FET) **512**.

[0029] In each output switching block **504**, the bootstrap circuitry **510** includes a bootstrap capacitor **522** and a switching device embodied by a PMOS FET **524** and bootstrap switching device driver circuitry **526**. The bootstrap capacitor **522** of each output switching block **504** is electrically coupled between the internal node **214** of the output switching block **504** and the bootstrap power supply rail **220** of the output switching block **504**. Additionally, in each output switching block **504**, the PMOS FET **524** is electrically coupled between the auxiliary power supply rail **126** and the bootstrap power supply rail **220** of the output switching block **504**, and the bootstrap switching device driver circuitry **526** drives a gate of the PMOS FET **524** in response to a signal bc generated by the controller **106** (FIG. 1). The bootstrap switching device driver circuitry **526** of each output switching block **504** is (i) powered from the bootstrap power supply rail **220** of the output switching block **504** and (ii) is electrically referenced to the internal node **214** of the output switching block **504**. As such, all switching device driver circuitry of a given output switching block **504** operates in a common power supply domain, i.e., between the bootstrap power supply rail **220** of the output switching block **504** and the internal node **214** of the output switching block **504**.

[0030] A drain of the reference NMOS FET **512** in each output switching block **504** is electrically coupled to the internal node **214** of the output switching block **504**, and the source of the reference NMOS FET **512** in each output switching block **504** is electrically coupled to the reference node **114**. The gate of each reference NMOS FET **512** is driven by a respective control signal rc generated by the controller **106** (FIG. 1).

[0031] Referring again to FIG. 1, the controller **106** is configured to generate the control signals for the power conversion stage **102** as well as the control signals for each output switching block **104**. Connections between the controller **106** and the power conversion stage **102**, as well as connections between the controller **106** and the output switching blocks **104**, are not shown for illustrative clarity. The controller **106** is implemented, for example, by analog and/or electronic circuitry. In some embodiments, the controller **106** is at least partially implemented by a processor (not shown) executing instructions in the form of software and/or firmware stored in a memory (not shown). Although the controller **106** is depicted as a discrete element for illustrative simplicity, the controller **106** could be partially or fully integrated with one or more other elements of the SIMO DC-to-DC converter **100**. For example, some subsystems of the controller **106** could be incorporated in the power conversion stage **102** and/or in one or more of the output switching blocks **104**. Additionally, the controller **106** may include multiple constituent elements that need not be co-packaged or even disposed at a common location.

[0032] As mentioned above, the controller **106** is configured to generate the control signal $pc1$, the control signal $pc2$, and/or the control signal $pc3$ to cause the power conversion stage **102** to provide a regulated voltage V_s with respect to the reference node **114** at the switching node **110**. The power conversion stage **102** has a buck and boost topology, and the magnitude of the voltage V_s may therefore be greater than the magnitude of the voltage V_{in} , less than a magnitude of the voltage V_{in} , or equal to magnitude of the

voltage V_m . For example, the controller **106** may cause the power conversion stage **102** to operate as a boost converter such that the magnitude of voltage V_s is greater than the magnitude of the voltage V_m by (i) generating the control signal pc1 to cause the power conversion switching device **116** to continuously operate in its on-state, (ii) generating the control signal pc2 to cause the power conversion switching device **118** to continuously operate in its off-state, and (iii) generating the control signal pc3 to modulate a duty cycle of the power conversion switching device **120** to achieve a desired magnitude of the voltage V_s . The duty cycle of a switching device is a portion of each switching cycle of the switching device that the switching device operates in its on-state. For example, if the power conversion switching device **120** operates in its on-state for 40 percent of its switching period, the power conversion switching device **120** has a duty cycle of 40 percent. In this document, a transistor or other switching device is in its on-state when the transistor or other switching device is being controlled to operate in its conductive state. Conversely, a transistor or other switching device is in its off-state when the transistor or other switching device is being controlled to operate in its non-conductive state.

[0033] FIG. 8 includes graphs **800**, **802**, and **804** collectively illustrating one example of the control signals pc1, pc2, and pc3 generated by the controller **106** when the controller **106** is causing the power conversion stage **102** to act as a boost converter such that the magnitude of the voltage V_s is greater than the magnitude of the voltage V_m . Graph **800** is of the control signal pc1 versus time, graph **802** is of the control signal pc2 versus time, and graph **804** is of the control signal pc3 versus time. Graphs **800**, **802**, and **804** have a common time base. FIG. 8 assumes that each of the power conversion switching device **116**, the power conversion switching device **118**, and the power conversion switching device **120** (i) operates in its on-state when its respective control signal is logic high and (ii) operates in its off-state when its respective control signal is logic low. However, the power conversion switching devices of power conversion stage **102** could be configured to respond to control signals in other manners without departing from the scope hereof. In the FIG. 8 example, the control signal pc1 is continuously logic high such that the power conversion switching device **116** continuously operates in its on-state, and the control signal pc2 is continuously logic low such that the power conversion switching device **118** continuously operates in its off-state. Additionally, the controller **106** generates the control signal pc3 in the FIG. 8 example such that (i) the control signal pc3 has a switching period T_a , (ii) the control signal pc3 is logic high for a duration of t_{on_a} of each switching period T_a , and (iii) the duration of t_{on_a} is 40 percent of the switching period T_a . As such, the power conversion switching device **120** has a duty cycle of 40 percent in the FIG. 8 example.

[0034] Referring again to FIG. 1, controller **106** may cause the power conversion stage **102** to operate as a buck converter such that the magnitude of the voltage V_s is less than or equal to the magnitude of the voltage V_m by (i) generating the control signal pc1 to modulate a duty cycle of the power conversion switching device **116** to achieve a desired magnitude of voltage V_s , (ii) generating the control signal pc2 to cause the power conversion switching device **118** to perform a freewheeling function, i.e., to provide a path for the current I_L flowing through the inductor **122** when the power con-

version switching device **116** is in its off-state, and (iii) generating the control signal pc3 to cause the power conversion switching device **120** to continuously operate in its off-state. For example, FIG. 9 includes graphs **900**, **902**, and **904** collectively illustrating one example of control signals pc1, pc2, and pc3 generated by the controller **106** when the controller **106** is causing the power conversion stage **102** to act as a buck converter such that the magnitude of the voltage V_s is less than the magnitude of the voltage V_m . Graph **900** is of the control signal pc1 versus time, graph **902** is of the control signal pc2 versus time, and graph **904** is of the control signal pc3 versus time. The graphs **900**, **902**, and **904** have a common time base. FIG. 9 assumes that each of the power conversion switching device **116**, the power conversion switching device **118**, and the power conversion switching device **120** (i) operates in its on-state when its respective control signal is logic high and (ii) operates in its off-state when its respective control signal is logic low. FIG. 9 also assumes that the current I_L through the inductor **122** is continuous.

[0035] In the FIG. 9 example, the controller **106** generates the control signal pc1 in the FIG. 9 example such that (i) the control signal pc1 has a switching period T_b , (ii) the control signal pc1 is logic high for a duration of t_{on_b} of each switching period T_b , and (iii) the duration of t_{on_b} is 60 percent of the switching period T_b . As such, the power conversion switching device **116** has a duty cycle of 60 percent in the FIG. 9 example. The control signal pc2 is complementary to the control signal pc1, and the control signal pc3 is continuously logic low, in the FIG. 9 example. Certain embodiments of the controller **106** are configured to insert deadtime between de-assertion of control signal pc1 and assertion of control signal pc2, and vice versa, to prevent shoot-through, i.e., simultaneous conduction of each of the power conversion switching device **116** and the power conversion switching device **118**.

[0036] Referring again to FIG. 1, the controller **106** may vary operation of the power conversion stage **102** according to which output power node **112** is being powered by the SIMO DC-to-DC converter **100**. For example, assume that (i) the SIMO DC-to-DC converter **100** is powering output power node **112(1)** and (ii) a load powered from the output power node **112(1)** requires a power supply voltage having a magnitude that is greater than the magnitude of the input voltage V_m . In this scenario, the controller **106** may cause the power conversion stage **102** to operate as a boost converter such that the magnitude of the voltage V_s is equal to the magnitude of the voltage required by the load powered from the output power node **112(1)**. Now assume that (i) the SIMO DC-to-DC converter **100** changes its operating state so that the SIMO DC-to-DC converter **100** is powering the output power node **112(2)** and (ii) a load powered from the output power node **112(1)** requires a power supply voltage having a magnitude that is smaller than the magnitude of the input voltage V_m . The controller **106** may accordingly cause the power conversion stage **102** to change from operating as a boost converter to operating as a buck converter such that the magnitude of the voltage V_s is equal to the magnitude of the voltage required by the load powered from the output power node **112(2)**. In some embodiments, the SIMO DC-to-DC converter **100** changes its active output power node **112** at a high rate, and the controller **106** may therefore cause the power conversion stage **102** to change its operating state at a corresponding high rate.

[0037] The controller 106 is configured to generate the control signals to control the output switching blocks 104 such that only one output power node 112 is powered by the SIMO DC-to-DC converter 100 at a given time, although some embodiments of the SIMO DC-to-DC converter 100 also support an alternate operating mode where two or more of the output power nodes 112 having a common voltage V_{out} may be simultaneously powered. The controller 106 is also configured to control the output switching blocks 104 so that at least one output switching block 104 provides a path for the current I_L flowing through the inductor 122 when (i) a magnitude of the current I_L is greater than zero and (ii) the power conversion switching device 120 is in its off-state. An output switching block 104 may provide a path for the current I_L by providing a path for the current I_L to flow to a load connected to the respective output power node 112 of the output switching block 104 via the input NMOS FET 202 and output NMOS FET 204 of the output switching block 104. Alternately, an output switching block 104 may provide a path for the current I_L to flow to the reference node 114 via the reference switching device 212 of the output switching block 104. Certain embodiments of the controller 106 are configured so that either the power conversion switching device 120 or at least one instance of an input NMOS FET 202 is in its on-state at any given time to ensure that there is path for the current I_L , irrespective of direction of flow of the current.

[0038] Additionally, in particular embodiments, the controller 106 is further configured such that an output NMOS FET 204 of a given output switching block 104 may operate in its on-state only if the power conversion switching device 120 is in its off-state and all other output NMOS FETs 204 are in their respective off-states. Furthermore, in some embodiments, the controller 106 is additionally configured such that (i) the power conversion switching device 120 may operate in its on-state only if all output NMOS FETs 204 are in their respective off-states and (ii) the power conversion switching device 120 may switch from its on-state to its off-state only once the PMOS FET 524 and the reference NMOS FET 512 of an output switching block 504 most-recently powering its respective output power node 112 have transitioned from their respective on-states to their respective off-states. Moreover, in certain embodiments, the controller 106 is further configured such to cause each of the PMOS FET 524 and the reference NMOS FET 512 of a given output switching block 504 that is powering its respective output power node 112 to remain in their respective on-states for a time window enclosed within the turn-on time of the power conversion switching device 120 in case the related output switching block 504 has its input NMOS FET 202 turned on and its output NMOS FET 204 turned off, or continuously as long as the related output switching block 504 has its input NMOS FET 202 and its output NMOS FET 204 both turned off.

[0039] FIG. 10 is a state diagram 1000 illustrating several possible operating states of each output switching block 104, although it is understood that the output switching blocks 104 could operate in other operating states without departing from the scope hereof. FIG. 10 assumes that the output switching blocks 104 are embodied as output switching blocks 504 and that (i) each switching device of the SIMO DC-to-DC converter 100 operates in its on-state with its respective control signal is logic high and (ii) each switching device of the SIMO DC-to-DC converter 100 operates in its

off-state with its respective control signal is logic low. State diagram 1000 illustrates three possible operating states of an output switching block 504, i.e., an on-state 1002, a first off-state 1004, and a second off-state 1006. The on-state 1002 of an output switching block 504 is characterized by the output switching block 504 electrically coupling its respective output power node 112 to the switching node 110, such that the respective output power node 112 is powered by the SIMO DC-to-DC converter 100. Accordingly, the control signal sbin and the control signal sbout of the output switching block 504 are logic high in the on-state 1002, to cause each of the input NMOS FET 202 and the output NMOS FET 204 of the output switching block 504 to operate in its on-state. Control signal rc of the output switching block 504 is logic low in the on-state 1002 to cause the reference NMOS FET 512 of the output switching block 504 to operate in its off-state. Additionally, the control signal bc of the output switching block 504 is logic low to cause the PMOS transistor 524 of the output switching block 504 to operate in its off-state, such that the bootstrap power supply rail 220 of the output switching block 504 is isolated from auxiliary power supply rail 126.

[0040] The first off-state 1004 of an output switching block 504 is characterized by the output switching block 504 isolating its respective output power node 112 from switching node 110, as well as by the output switching block 504 isolating its respective output power node 112 from the other output power nodes 112. Therefore, the respective output power node 112 of the output switching block 504 is not powered by the SIMO DC-to-DC converter 100 when the output switching block 504 is in the first off-state 1004. Accordingly, the control signal sbin and the control signal sbout of the output switching block 504 are logic low in the first off-state 1004, to cause each of the input NMOS FET 202 and the output NMOS FET 204 of the output switching block 504 to operate in its off-state. The control signal rc of the output switching block 504 is logic high in the first off-state 1004 to cause the reference NMOS FET 512 of the output switching block 504 to operate in its on-state. Additionally, the control signal bc of the output switching block 504 is logic high in the first off-state 1004 to cause the PMOS transistor 524 of the output switching block 504 to operate in its on-state. Consequently, the bootstrap capacitor 522 of the output switching block 504 charges from the auxiliary power supply rail 126 and through the PMOS FET 524 of the output switching block 504, in the first off-state 1004 of the output switching block 504.

[0041] The second off-state 1006 of an output switching block 504 is characterized by the SIMO DC-to-DC converter 100 not powering the respective output power node 112 of the output switching block 504. Accordingly, control signal sbout of the output switching block 504 is logic low in the second off-state 1006 to cause the output NMOS FET 204 of the output switching block 504 to operate in its off-state. However, control signal sbin of the output switching block 504 is logic high in the second off-state 1006 to cause the first input NMOS FET 202 of the output switching block 504 to operate in its on-state. Therefore, during dead times, current may flow from switching node 110 to the respective output power node 112 of the output switching block 504 via the body diode 218 of the output NMOS FET 204 of the output switching block 504. Additionally, control signal rc of the output switching block 504 is logic high in the second off-state 1006 to cause the reference NMOS FET

512 to operate in its on-state. Therefore, the output switching block **504** provides a path for current I_L to flow through the reference NMOS FET **512** of the output switching block **504** in the second off-state **1006**. Control signal **bc** of the output switching block **504** is also logic high in the second off-state **1006** to cause the PMOS transistor **524** of the output switching block **504** to operate in its on-state. Consequently, the bootstrap capacitor **522** of the output switching block **504** charges from the auxiliary power supply rail **126** and through the PMOS FET **524** of the output switching block **504** in the second off-state **1006** of the output switching block **504**. The controller **106** causes the output switching block **504** to operate in the second off-state **1006** instead of the first off-state **1004**, for example, if the output switching block **504** is expected to quickly transition back to the on-state **1002**, such as when the output NMOS FET **204** of the output switching block **504** needs to switch between its on-state and its off-state multiple times without the output NMOS FETs **204** of other output switching blocks **504** needing to operate in their respective on-states in the meantime. Operation in the second off-state **1006** instead of in the first off-state **1004** is this scenario advantageously eliminates losses caused by unnecessary switching of the input NMOS FET **202**.

[0042] FIG. 11 includes graphs **1100**, **1102**, **1104**, **1106**, and **1108** collectively illustrating an example transition **1008** (FIG. 10) of an output switching block **504** from the first off-state **1004** to the on-state **1002**. Graph **1100** is of the control signal **pc3** versus time, graph **1102** is of the control signal **sbin** versus time, graph **1104** is of the control signal **sbout** versus time, graph **1106** is of the control signal **rc** versus time, and graph **1108** is of the control signal **bc** versus time. Graphs **1100**, **1102**, **1104**, **1106**, and **1108** have a common time base. The output switching block **504** transitions from the first off-state **1004** to the on-state **1002** at a time t_1 in the FIG. 11 example. The output switching block **504** accordingly operates in the first off-state **1004** before the time t_1 , wherein each of the control signal **sbin** and the control signal **sbout** is logic low, and each of the control signal **rc** and the control signal **bc** is logic high. Additionally, the control signal **pc3** is logic high during the first off-state **1004** in the FIG. 11 example. The output switching block **504** transitions from the first off-state **1004** to the on-state **1002** at the time t_1 , and each of the control signal **rc** and the control signal **bc** transitions from logic high to logic low at time t_1 . The control signal **sbin** next transitions from logic low to logic high, followed by the control signal **pc3** transitioning from logic high to logic low. Finally, the control signal **sbout** transitions from logic low to logic high after the control signal **pc3** has transitioned from logic high to logic low. While power conversion switching device **120** can no longer provide a path for the current I_L after the control signal **pc3** transitions to logic low in the on-state **1002**, the output switching block **504** provides a path for the current I_L to a load powered from its respective output power node **112** during the on-state **1002**.

[0043] FIG. 12 includes graphs **1200**, **1202**, **1204**, **1206**, and **1208** collectively illustrating an example transition **1010** (FIG. 10) of an output switching block **504** from the on-state **1002** to the first off-state **1004**. Graph **1200** is of the control signal **pc3** versus time, graph **1202** is of the control signal **sbin** versus time, graph **1204** is of the control signal **sbout** versus time, graph **1206** is of the control signal **rc** versus time, and graph **1208** is of the control signal **bc** versus time.

Graphs **1200**, **1202**, **1204**, **1206**, and **1208** have a common time base. The output switching block **504** transitions from the on-state **1002** to the first off-state **1004** at a time t_2 in the FIG. 12 example. The output switching block **504** accordingly operates in the on-state **1002** before the time t_2 , wherein each of the control signal **sbin** and the control signal **sbout** is logic high, and each of the control signal **rc** and the control signal **bc** is logic low. Additionally, the control signal **pc3** is logic low during the on-state **1002** in the FIG. 12 example. The output switching block **504** transitions from the on-state **1002** to the first off-state **1004** at the time t_2 , and the control signal **sbout** accordingly transitions from logic high to logic low, followed by the control signal **pc3** transitioning from logic low to logic high. The control signal **sbin** transitions from logic high to logic low after control signal **pc3** transitions to logic high. Finally, the control signal **rc** and the control signal **bc** transition from logic low to logic high together after the control signal **sbin** has transitioned from logic high to logic low. Once the power conversion switching device **120** is in its on-state, the output switching block **504** no longer needs to provide a path for the current I_L .

[0044] FIG. 13 includes graphs **1300**, **1302**, **1304**, **1306**, and **1308** collectively illustrating an example transition **1012** (FIG. 10) of an output switching block **504** from the second off-state **1006** to the on-state **1002**. Graph **1300** is of the control signal **pc3** versus time, graph **1302** is of the control signal **sbin** versus time, graph **1304** is of the control signal **sbout** versus time, graph **1306** is of the control signal **rc** versus time, and graph **1308** is of the control signal **bc** versus time. Graphs **1300**, **1302**, **1304**, **1306**, and **1308** have a common time base. The output switching block **504** transitions from the second off-state **1006** to the on-state **1002** at a time t_3 in the FIG. 13 example. The output switching block **504** accordingly operates in the second off-state **1006** before the time t_3 , wherein the control signal **sbin** is logic high, the control signal **sbout** is logic low, and each of the control signal **rc** and the control signal **bc** is logic high. Additionally, the control signal **pc3** is logic high during the second off-state **1006** in the FIG. 13 example. The output switching block **504** transitions from the second off-state **1006** to the on-state **1002** at the time t_3 , and the control signal **rc** and the control signal **bc** transition from logic high to logic low together. Next, the control signal **pc3** transitions from logic high to logic low after each of the control signal **rc** and the control signal **bc** have transitioned from logic high to logic low, and the control signal **sbout** transitions from logic low to logic high after the control signal **pc3** has transitioned to logic low. The output switching block **504** provides a path for the current I_L to a load powered from its respective output power node **112** once the control signal **sbout** is logic high.

[0045] FIG. 14 includes graphs **1400**, **1402**, **1404**, **1406**, and **1408** collectively illustrating an example transition **1014** (FIG. 10) of an output switching block **504** from the on-state **1002** to the second off-state **1006**. Graph **1400** is of the control signal **pc3** versus time, graph **1402** is of the control signal **sbin** versus time, graph **1404** is of the control signal **sbout** versus time, graph **1406** is of the control signal **rc** versus time, and graph **1408** is of the control signal **bc** versus time. Graphs **1400**, **1402**, **1404**, **1406**, and **1408** have a common time base. The output switching block **504** transitions from the on-state **1002** to the second off-state **1006** at a time t_4 in the FIG. 14 example. The output switching block

504 accordingly operates in the on-state **1002** before the time t_4 , wherein each of the control signal sbin and the control signal sbout is logic high, and each of the control signal rc and the control signal bc is logic low. Additionally, the control signal pc3 is logic low during the on-state **1002** in the FIG. 14 example. The output switching block **504** transitions from the on-state **1002** to the second off-state **1006** at the time t_4 , and the control signal sbout accordingly transitions from logic high to logic low. The control signal pc3 transitions from logic low to logic high after the control signal sbout is logic low. Finally, the control signal rc and the control signal bc transition from logic low to logic high together after the control signal pc3 is logic high. Consequently, there is a path for the current I_L through both of the power conversion switching device **120** and the reference NMOS FET **512** of the output switching block **504** in the second off-state **1006** after switching transitions are complete slightly after time t_4 .

[0046] Referring again to FIG. 5-7, any of the output switching blocks **504** could be modified to include one or more additional elements. For example, FIG. 15 is a schematic diagram of an output switching block **1504(1)**, which is an alternate embodiment of the output switching block **504(1)** (FIG. 5) further including elements to perform current sensing. It is understood that the other output switching blocks **504** could be modified in a manner similar to that illustrated in FIG. 15. The output switching block **1504(1)** differs from the output switching block **504(1)** in that output switching block **1504(1)** further includes first replica circuitry **1528**, second replica circuitry **1530**, and a current sense comparator **1532**. Each of the first replica circuitry **1528** and the second replica circuitry **1530** is a scaled model of the output NMOS FET **204(1)**. Additionally, each of the first replica circuitry **1528** and the second replica circuitry **1530** operates in the same power supply domain as the switching device driver circuitry of the output switching block **1504(1)**, i.e., between the bootstrap power supply rail **220(1)** and the internal node **214(1)**. The first replica circuitry **1528** generates a signal rs1 in response to voltage at the internal node **214(1)**, and the second replica circuitry **1530** generates a signal rs2 in response to voltage at the output power node **112(1)**. The current sense comparator **1532** compares the signal rs1 and the signal rs2 to generate a signal i_s representing magnitude of current flowing through the output NMOS FET **204(1)**.

Combinations of Features

[0047] Features described above may be combined in various ways without departing from the scope hereof. The following examples illustrate some possible combinations.

[0048] (A1) A single-input, multiple-output (SIMO) direct-current-to-direct-current (DC-to-DC) converter includes a power conversion stage and a plurality of output switching blocks. The power conversion stage is electrically coupled between an input power node and a switching node, and the power conversion stage includes a first power conversion switching device electrically coupled between the switching node and a reference node. Each output switching block is electrically coupled between the switching node and a respective output power node. Each output switching block includes (1) a respective input enhancement mode, N-type metal oxide semiconductor field effect transistor (input NMOS FET) of the output switching block that is electrically coupled between the switching node and a

respective internal node of the output switching block, where the input NMOS FET includes a respective body diode oriented such that (i) a cathode of the body diode is electrically coupled to the switching node and (ii) an anode of the body diode is electrically coupled to the respective internal node of the output switching block, and (2) a respective output enhancement mode, N-type metal oxide semiconductor field effect transistor (output NMOS FET) of the output switching block that is electrically coupled between the respective internal node of the output switching block and the respective output power node of the output switching block, where the output NMOS FET includes a respective body diode oriented such that (i) a cathode of the body diode is electrically coupled to the respective output power node of the output switching block and (ii) an anode of the body diode is electrically coupled to the respective internal node of the output switching block.

[0049] (A2) In the SIMO DC-to-DC converter denoted as (A1), each output switching block may further include (1) respective bootstrap circuitry of the output switching block that is configured to generate a respective bootstrap power supply rail of the output switching block referenced to the respective internal node of the output switching block, (2) respective input switch driver circuitry of the output switching block that is configured to drive a gate of the respective input NMOS FET of the output switching block and is electrically powered by the respective bootstrap power supply rail of the output switching block, and (3) respective output switch driver circuitry of the output switching block that is configured to drive a gate of the respective output NMOS FET of the output switching block and is electrically powered by the respective bootstrap power supply rail of the output switching block.

[0050] (A3) The SIMO DC-to-DC converter denoted as (A1) may further include a controller configured to control at least each of the first power conversion switching device, the respective input NMOS FET of each output switching block, and the respective output NMOS FET of each output switching block.

[0051] (A4) In the SIMO DC-to-DC converter denoted as (A3), the controller may be further configured to control each of the first power conversion switching device, the respective input NMOS FET of each output switching block, and the respective output NMOS FET of each output switching block in a manner which provides a path for current flowing through an inductor of the power conversion stage at any time during operation of the SIMO DC-to-DC converter.

[0052] (A5) In either one of the SIMO DC-to-DC converters denoted as (A3) or (A4), the controller may be further configured to control the first power conversion switching device such that the first power conversion switching device operates in its on-state solely when the respective output NMOS FET of each output switching block operates in its respective off-state.

[0053] (A6) In any one of the SIMO DC-to-DC converters denoted as (A1) through (A5), (i) each output switching block may further include a respective reference switching device electrically coupled between the respective internal node of the output switching block and the reference node, (ii) each output switching block may further include a respective bootstrap capacitor and a respective bootstrap control switching device electrically coupled in series

between the respective internal node of the output switching block and an auxiliary power supply rail.

[0054] (A7) In the SIMO DC-to-DC converter denoted as (A6), the controller may be further configured, with respect to each output switching block, to control the first power conversion switching device to cause the first power conversion switching device to switch from its on-state to its off-state after each of (i) the respective reference switching device of the output switching block has switched from its on-state to its off-state and (ii) the respective bootstrap control switching device of the output switching block has switched from its on-state to its off-state.

[0055] (A8) In either one of the SIMO DC-to-DC converters denoted as (A6) or (A7), the controller may be further configured to control the respective reference switching device of each output switching block and the respective bootstrap control switching device of each output switching block such that in a given output switching block, the respective reference switching device of the output switching block and the respective bootstrap control switching device of the output switching block do not switch from their respective off-states to their respective on-state until after (i) the respective output NMOS FET of the output switching block has switched from its on-state to its off-state and (ii) the first power conversion switching device has switched from its off-state to its on-state.

[0056] (A9) In any one of the SIMO DC-to-DC converters denoted as (A3) through (A8), wherein the controller may be further configured to control the respective output NMOS FET of each output switching block in a manner such that at most one output NMOS FET is in its on-state at any given time during operation of the SIMO DC-to-DC converter.

[0057] (A10) In any one of the SIMO DC-to-DC converters denoted as (A3) through (A9), the controller may be further configured to control the respective output NMOS FET of each output switching block such that each output NMOS FET does not switch from its off-state to its on-state until after the first power conversion device switches from its on-state to its off-state.

[0058] (A11) In any one of the SIMO DC-to-DC converters denoted as (A3) through (A10), the controller may be further configured to control the respective input NMOS FET of each output switching block such that in a given output switching block, the respective input NMOS FET of the output switching block switches from its off-state to its on-state before the first power conversion switching device switches from its on-state to its off-state.

[0059] (A12) In any one of the SIMO DC-to-DC converters denoted as (A3) through (A11), the controller may be further configured to control the respective input NMOS FET of each output switching block such that in a given output switching block, the respective input NMOS FET of the output switching block switches from its on-state to its off-state after the first power conversion switching device switches from its off-state to its on-state.

[0060] (A13) In the SIMO DC-to-DC converter denoted as (A2), the respective bootstrap circuitry of each output switching block may include (i) a respective bootstrap capacitor electrically coupled between the respective bootstrap power supply rail of the output switching block and the respective internal node of the output switching block and (ii) a respective bootstrap control switching device electrically coupled between an auxiliary power supply rail and the respective bootstrap capacitor of the output switching block.

[0061] (A14) In any one of the SIMO DC-to-DC converters denoted as (A1) through (A13), the power conversion stage may have a buck and boost topology.

[0062] (B1) A single-input, multiple-output (SIMO) direct-current-to-direct-current (DC-to-DC) converter includes a power conversion stage, a first output switching block, and a second output switching block. The power conversion stage electrically is coupled between an input power node and a switching node, and the power conversion stage includes a first power conversion switching device electrically coupled between the switching node and a reference node. The first output switching block includes (1) a first input enhancement mode, N-type metal oxide semiconductor field effect transistor (first input NMOS FET) electrically coupled between the switching node and a first internal node, where the first input NMOS FET includes a body diode oriented such that (i) a cathode of the body diode is electrically coupled to the switching node and (ii) an anode of the body diode is electrically coupled to the first internal node, and (2) a first output enhancement mode, N-type metal oxide semiconductor field effect transistor (first output NMOS FET) electrically coupled between the first internal node and a first output power node, where the first output NMOS FET includes a body diode oriented such that (i) a cathode of the body diode is electrically coupled to the first output power node and (ii) an anode of the body diode is electrically coupled to the first internal node. The second output switching block includes (1) a second input enhancement mode, N-type metal oxide semiconductor field effect transistor (second input NMOS FET) electrically coupled between the switching node and a second internal node, where the second input NMOS FET includes a body diode oriented such that (i) a cathode of the body diode is electrically coupled to the switching node and (ii) an anode of the body diode is electrically coupled to the second internal node, and (2) a second output enhancement mode, N-type metal oxide semiconductor field effect transistor (second output NMOS FET) electrically coupled between the second internal node and a second output power node, where the second output NMOS FET includes a body diode oriented such that (i) a cathode of the body diode is electrically coupled to the second output power node and (ii) an anode of the body diode is electrically coupled to the second internal node.

[0063] (B2) In the SIMO DC-to-DC converter denoted as (B1), (1) the first output switching block may further include (i) first bootstrap circuitry configured to generate a first bootstrap power supply rail referenced to the first internal node, (ii) first input switch driver circuitry configured to drive a gate of the first input NMOS FET and being electrically powered by the first bootstrap power supply rail, and (iii) first output switch driver circuitry configured to drive a gate of the first output NMOS FET and being electrically powered by the first bootstrap power supply rail, and (2) the second output switching block may further include (i) second bootstrap circuitry configured to generate a second bootstrap power supply rail referenced to the second internal node, (ii) second input switch driver circuitry configured to drive a gate of the second input NMOS FET and being electrically powered by the second bootstrap power supply rail, and (iii) second output switch driver circuitry configured to drive a gate of the second output NMOS FET and being electrically powered by the second bootstrap power supply rail.

[0064] (B3) Either one of the SIMO DC-to-DC converters denoted as (B1) and (B2) may further include a controller configured to control at least each of the first power conversion switching device, the first input NMOS FET, the first output NMOS FET, the second input NMOS FET, and the second output NMOS FET.

[0065] (B4) In the SIMO DC-to-DC converter denoted as (B3), the controller may be further configured to control at least each of the first power conversion switching device, the first input NMOS FET, the first output NMOS FET, the second input NMOS FET, and the second output NMOS FET in a manner which provides a path for current flowing through an inductor of the power conversion stage at any time during operation of the SIMO DC-to-DC converter.

[0066] (B5) Any one of the SIMO DC-to-DC converters denoted as (B1) through (B4) may further include a third output switching block including (1) a third input enhancement mode, N-type metal oxide semiconductor field effect transistor (third input NMOS FET) electrically coupled between the switching node and a third internal node, where the third input NMOS FET includes a body diode oriented such that (i) a cathode of the body diode is electrically coupled to the switching node and (ii) an anode of the body diode is electrically coupled to the third internal node, and (2) a third output enhancement mode, N-type metal oxide semiconductor field effect transistor (third output NMOS FET) electrically coupled between the third internal node and a third output power node, where the third output NMOS FET includes a body diode oriented such that (i) a cathode of the body diode is electrically coupled to the third output power node and (ii) an anode of the body diode is electrically coupled to the third internal node.

[0067] (C1) A method for controlling a plurality of output switching blocks in a single-input, multiple-output (SIMO) direct-current-to-direct-current (DC-to-DC) converter includes, in each of the plurality of output switching blocks, (1) generating a respective bootstrap power supply rail of the output switching block referenced to a respective internal node of the output switching block, (2) driving a gate of a respective input enhancement mode, N-type metal oxide semiconductor field effect transistor (input NMOS FET) of the output switching block between a voltage of the respective bootstrap power supply rail of the output switching block and a voltage of the respective internal node of the output switching block, where a drain of the respective input NMOS FET of the output switching block is electrically coupled to a switching node of the SIMO DC-to-DC converter, and a source of the respective input NMOS FET of the output switching block is electrically coupled to the respective internal node of the output switching block, and (3) driving a gate of a respective output enhancement mode, N-type metal oxide semiconductor field effect transistor (output NMOS FET) of the output switching block between a voltage of the respective bootstrap power supply rail of the output switching block and a voltage of the respective internal node of the output switching block, where a drain of the respective output NMOS FET of the output switching block is electrically coupled to a respective output power node of the output switching block, and a source of the respective output NMOS FET of the output switching block is electrically coupled to the respective internal node of the output switching block.

[0068] (C2) The method denoted as (C1) may further include controlling the respective input NMOS FET of each

output switching block and the respective output NMOS FET of each output switching block such that the respective output power node of at most one output switching block at any given time is powered by the SIMO DC-to-DC converter during operation of the SIMO DC-to-DC converter.

[0069] Changes may be made in the above methods, devices, and systems without departing from the scope hereof. For example, power conversion stage 102 could be modified to have a different topology, such a buck topology or a boost topology instead of a buck and boost topology. It should thus be noted that the matter contained in the above description and shown in the accompanying drawings should be interpreted as illustrative and not in a limiting sense. The following claims are intended to cover generic and specific features described herein, as well as all statements of the scope of the present method and system, which as a matter of language, might be said to fall therebetween.

What is claimed is:

1. A single-input, multiple-output (SIMO) direct-current-to-direct-current (DC-to-DC) converter, comprising:

a power conversion stage electrically coupled between an input power node and a switching node, the power conversion stage including a first power conversion switching device electrically coupled between the switching node and a reference node; and

a plurality of output switching blocks, each output switching block being electrically coupled between the switching node and a respective output power node, each output switching block including:

a respective input enhancement mode, N-type metal oxide semiconductor field effect transistor (input NMOS FET) of the output switching block that is electrically coupled between the switching node and a respective internal node of the output switching block, the input NMOS FET including a respective body diode oriented such that (i) a cathode of the body diode is electrically coupled to the switching node and (ii) an anode of the body diode is electrically coupled to the respective internal node of the output switching block, and

a respective output enhancement mode, N-type metal oxide semiconductor field effect transistor (output NMOS FET) of the output switching block that is electrically coupled between the respective internal node of the output switching block and the respective output power node of the output switching block, the output NMOS FET including a respective body diode oriented such that (i) a cathode of the body diode is electrically coupled to the respective output power node of the output switching block and (ii) an anode of the body diode is electrically coupled to the respective internal node of the output switching block.

2. The SIMO DC-to-DC converter of claim 1, wherein each output switching block further includes:

respective bootstrap circuitry of the output switching block that is configured to generate a respective bootstrap power supply rail of the output switching block referenced to the respective internal node of the output switching block,

respective input switch driver circuitry of the output switching block that is configured to drive a gate of the respective input NMOS FET of the output switching

block and is electrically powered by the respective bootstrap power supply rail of the output switching block, and

respective output switch driver circuitry of the output switching block that is configured to drive a gate of the respective output NMOS FET of the output switching block and is electrically powered by the respective bootstrap power supply rail of the output switching block.

3. The SIMO DC-to-DC converter of claim 1, further comprising a controller configured to control at least each of the first power conversion switching device, the respective input NMOS FET of each output switching block, and the respective output NMOS FET of each output switching block.

4. The SIMO DC-to-DC converter of claim 3, wherein the controller is further configured to control each of the first power conversion switching device, the respective input NMOS FET of each output switching block, and the respective output NMOS FET of each output switching block in a manner which provides a path for current flowing through an inductor of the power conversion stage at any time during operation of the SIMO DC-to-DC converter.

5. The SIMO DC-to-DC converter of claim 3, wherein the controller is further configured to control the first power conversion switching device such that the first power conversion switching device operates in its on-state solely when the respective output NMOS FET of each output switching block operates in its respective off-state.

6. The SIMO DC-to-DC converter of claim 3, wherein: each output switching block further includes a respective reference switching device electrically coupled between the respective internal node of the output switching block and the reference node;

each output switching block further includes a respective bootstrap capacitor and a respective bootstrap control switching device electrically coupled in series between the respective internal node of the output switching block and an auxiliary power supply rail; and

with respect to each output switching block, the controller is further configured to control the first power conversion switching device to cause the first power conversion switching device to switch from its on-state to its off-state after each of (i) the respective reference switching device of the output switching block has switched from its on-state to its off-state and (ii) the respective bootstrap control switching device of the output switching block has switched from its on-state to its off-state.

7. The SIMO DC-to-DC converter of claim 3, wherein the controller is further configured to control the respective output NMOS FET of each output switching block in a manner such that at most one output NMOS FET is in its on-state at any given time during operation of the SIMO DC-to-DC converter.

8. The SIMO DC-to-DC converter of claim 3, wherein the controller is further configured to control the respective output NMOS FET of each output switching block such that each output NMOS FET does not switch from its off-state to its on-state until after the first power conversion device switches from its on-state to its off-state.

9. The SIMO DC-to-DC converter of claim 3, wherein: each output switching block further includes a respective reference switching device electrically coupled

between the respective internal node of the output switching block and the reference node;

each output switching block further includes a respective bootstrap capacitor and a respective bootstrap control switching device electrically coupled in series between the respective internal node of the output switching block and an auxiliary power supply rail; and

the controller is further configured to control the respective reference switching device of each output switching block and the respective bootstrap control switching device of each output switching block such that in a given output switching block, the respective reference switching device of the output switching block and the respective bootstrap control switching device of the output switching block do not switch from their respective off-states to their respective on-state until after (i) the respective output NMOS FET of the output switching block has switched from its on-state to its off-state and (ii) the first power conversion switching device has switched from its off-state to its on-state.

10. The SIMO DC-to-DC converter of claim 3, wherein the controller is further configured to control the respective input NMOS FET of each output switching block such that in a given output switching block, the respective input NMOS FET of the output switching block switches from its off-state to its on-state before the first power conversion switching device switches from its on-state to its off-state.

11. The SIMO DC-to-DC converter of claim 3, wherein the controller is further configured to control the respective input NMOS FET of each output switching block such that in a given output switching block, the respective input NMOS FET of the output switching block switches from its on-state to its off-state after the first power conversion switching device switches from its off-state to its on-state.

12. The SIMO DC-to-DC converter of claim 2, wherein the respective bootstrap circuitry of each output switching block includes:

a respective bootstrap capacitor electrically coupled between the respective bootstrap power supply rail of the output switching block and the respective internal node of the output switching block; and

a respective bootstrap control switching device electrically coupled between an auxiliary power supply rail and the respective bootstrap capacitor of the output switching block.

13. The SIMO DC-to-DC converter of claim 1, wherein the power conversion stage has a buck and boost topology.

14. A single-input, multiple-output (SIMO) direct-current-to-direct-current (DC-to-DC) converter, comprising:

a power conversion stage electrically coupled between an input power node and a switching node, the power conversion stage including a first power conversion switching device electrically coupled between the switching node and a reference node;

a first output switching block including:

a first input enhancement mode, N-type metal oxide semiconductor field effect transistor (first input NMOS FET) electrically coupled between the switching node and a first internal node, the first input NMOS FET including a body diode oriented such that (i) a cathode of the body diode is electrically coupled to the switching node and (ii) an anode of the body diode is electrically coupled to the first internal node, and

- a first output enhancement mode, N-type metal oxide semiconductor field effect transistor (first output NMOS FET) electrically coupled between the first internal node and a first output power node, the first output NMOS FET including a body diode oriented such that (i) a cathode of the body diode is electrically coupled to the first output power node and (ii) an anode of the body diode is electrically coupled to the first internal node; and
 - a second output switching block including:
 - a second input enhancement mode, N-type metal oxide semiconductor field effect transistor (second input NMOS FET) electrically coupled between the switching node and a second internal node, the second input NMOS FET including a body diode oriented such that (i) a cathode of the body diode is electrically coupled to the switching node and (ii) an anode of the body diode is electrically coupled to the second internal node, and
 - a second output enhancement mode, N-type metal oxide semiconductor field effect transistor (second output NMOS FET) electrically coupled between the second internal node and a second output power node, the second output NMOS FET including a body diode oriented such that (i) a cathode of the body diode is electrically coupled to the second output power node and (ii) an anode of the body diode is electrically coupled to the second internal node.
15. The SIMO DC-to-DC converter of claim 14, wherein: the first output switching block further includes:
- first bootstrap circuitry configured to generate a first bootstrap power supply rail referenced to the first internal node,
 - first input switch driver circuitry configured to drive a gate of the first input NMOS FET and being electrically powered by the first bootstrap power supply rail, and
 - first output switch driver circuitry configured to drive a gate of the first output NMOS FET and being electrically powered by the first bootstrap power supply rail; and
- the second output switching block further includes:
- second bootstrap circuitry configured to generate a second bootstrap power supply rail referenced to the second internal node,
 - second input switch driver circuitry configured to drive a gate of the second input NMOS FET and being electrically powered by the second bootstrap power supply rail, and
 - second output switch driver circuitry configured to drive a gate of the second output NMOS FET and being electrically powered by the second bootstrap power supply rail.
16. The SIMO DC-to-DC converter of claim 14, further comprising a controller configured to control at least each of the first power conversion switching device, the first input NMOS FET, the first output NMOS FET, the second input NMOS FET, and the second output NMOS FET.
17. The SIMO DC-to-DC converter of claim 14, wherein the controller is further configured to control at least each of the first power conversion switching device, the first input NMOS FET, the first output NMOS FET, the second input NMOS FET, and the second output NMOS FET in a manner

which provides a path for current flowing through an inductor of the power conversion stage at any time during operation of the SIMO DC-to-DC converter.

18. The SIMO DC-to-DC converter of claim 14, further comprising a third output switching block including:

- a third input enhancement mode, N-type metal oxide semiconductor field effect transistor (third input NMOS FET) electrically coupled between the switching node and a third internal node, the third input NMOS FET including a body diode oriented such that (i) a cathode of the body diode is electrically coupled to the switching node and (ii) an anode of the body diode is electrically coupled to the third internal node; and
- a third output enhancement mode, N-type metal oxide semiconductor field effect transistor (third output NMOS FET) electrically coupled between the third internal node and a third output power node, the third output NMOS FET including a body diode oriented such that (i) a cathode of the body diode is electrically coupled to the third output power node and (ii) an anode of the body diode is electrically coupled to the third internal node.

19. A method for controlling a plurality of output switching blocks in a single-input, multiple-output (SIMO) direct-current-to-direct-current (DC-to-DC) converter, the method comprising, in each of the plurality of output switching blocks:

- generating a respective bootstrap power supply rail of the output switching block referenced to a respective internal node of the output switching block;
- driving a gate of a respective input enhancement mode, N-type metal oxide semiconductor field effect transistor (input NMOS FET) of the output switching block between a voltage of the respective bootstrap power supply rail of the output switching block and a voltage of the respective internal node of the output switching block, a drain of the respective input NMOS FET of the output switching block being electrically coupled to a switching node of the SIMO DC-to-DC converter, and a source of the respective input NMOS FET of the output switching block being electrically coupled to the respective internal node of the output switching block; and
- driving a gate of a respective output enhancement mode, N-type metal oxide semiconductor field effect transistor (output NMOS FET) of the output switching block between a voltage of the respective bootstrap power supply rail of the output switching block and a voltage of the respective internal node of the output switching block, a drain of the respective output NMOS FET of the output switching block being electrically coupled to a respective output power node of the output switching block, and a source of the respective output NMOS FET of the output switching block being electrically coupled to the respective internal node of the output switching block.

20. The method of claim 19, further comprising controlling the respective input NMOS FET of each output switching block and the respective output NMOS FET of each output switching block such that the respective output power node of at most one output switching block at any given time is powered by the SIMO DC-to-DC converter during operation of the SIMO DC-to-DC converter.